GROSS ANTOMY AND HISTOLOGY OF EYE

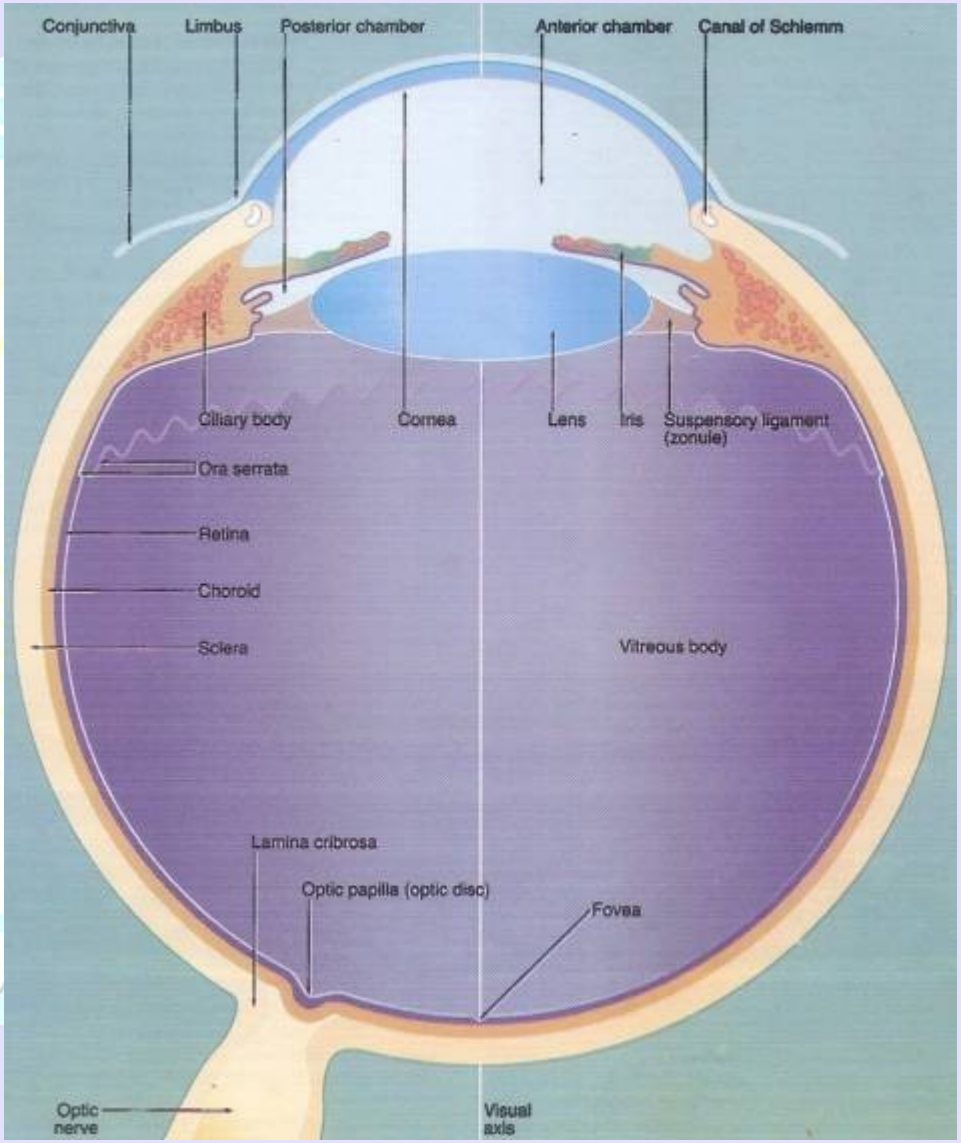
“LEARNING OBJECTIVE”

- **At the end of lecture student must be able to know**

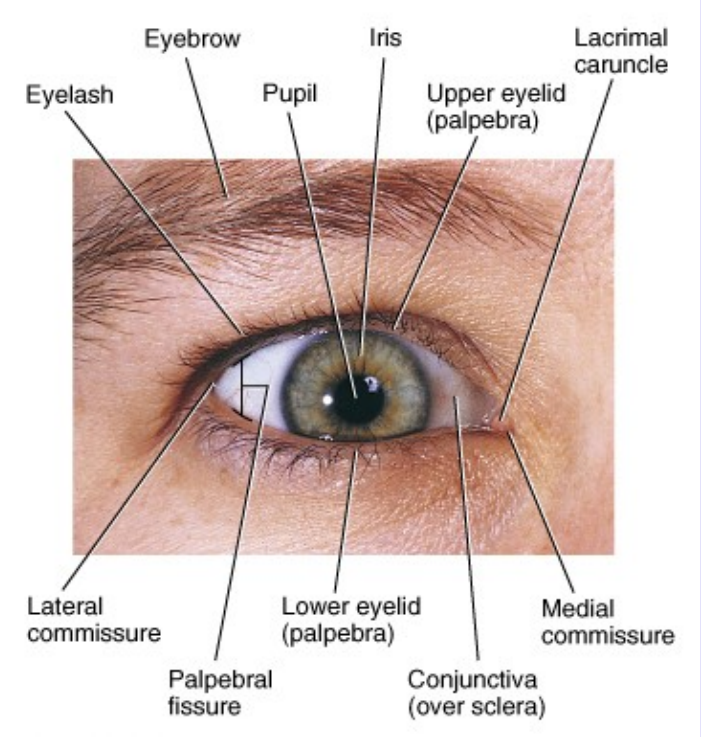
- The eye ball and its three coats.
- The details of sclera, cornea, uveal tract and retina and their microscopic picture.
- Layers of the retina
- The histology and functions of pigment epithelium
- The histology of photoreceptors, associated neurons, supporting elements and fovea centralis

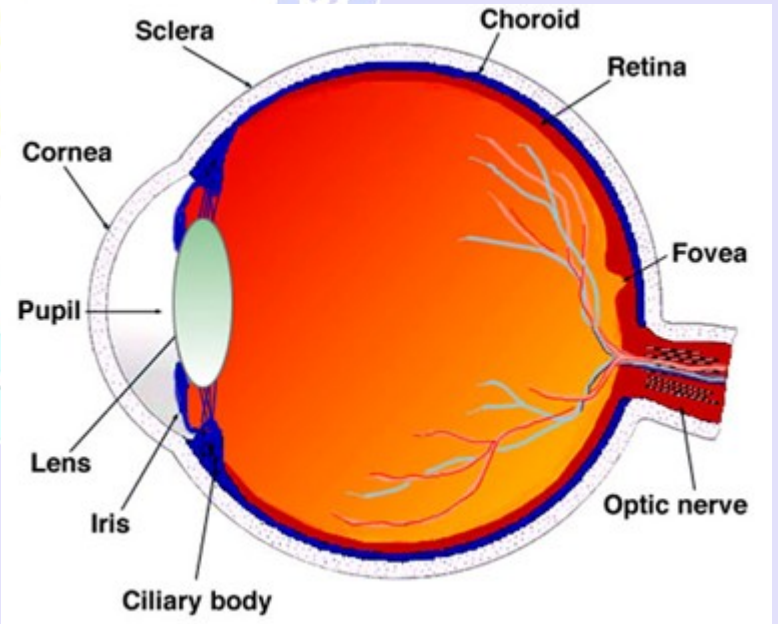
THE GROSS ANATOMY OF THE EYE

- Eye is like camera having,
- Light sensitive retina
- A lens system for focusing images (the cornea, lens and refractive media)
- Light controlling mechanism the iris diaphragm
- Inside is black to prevent internal reflection

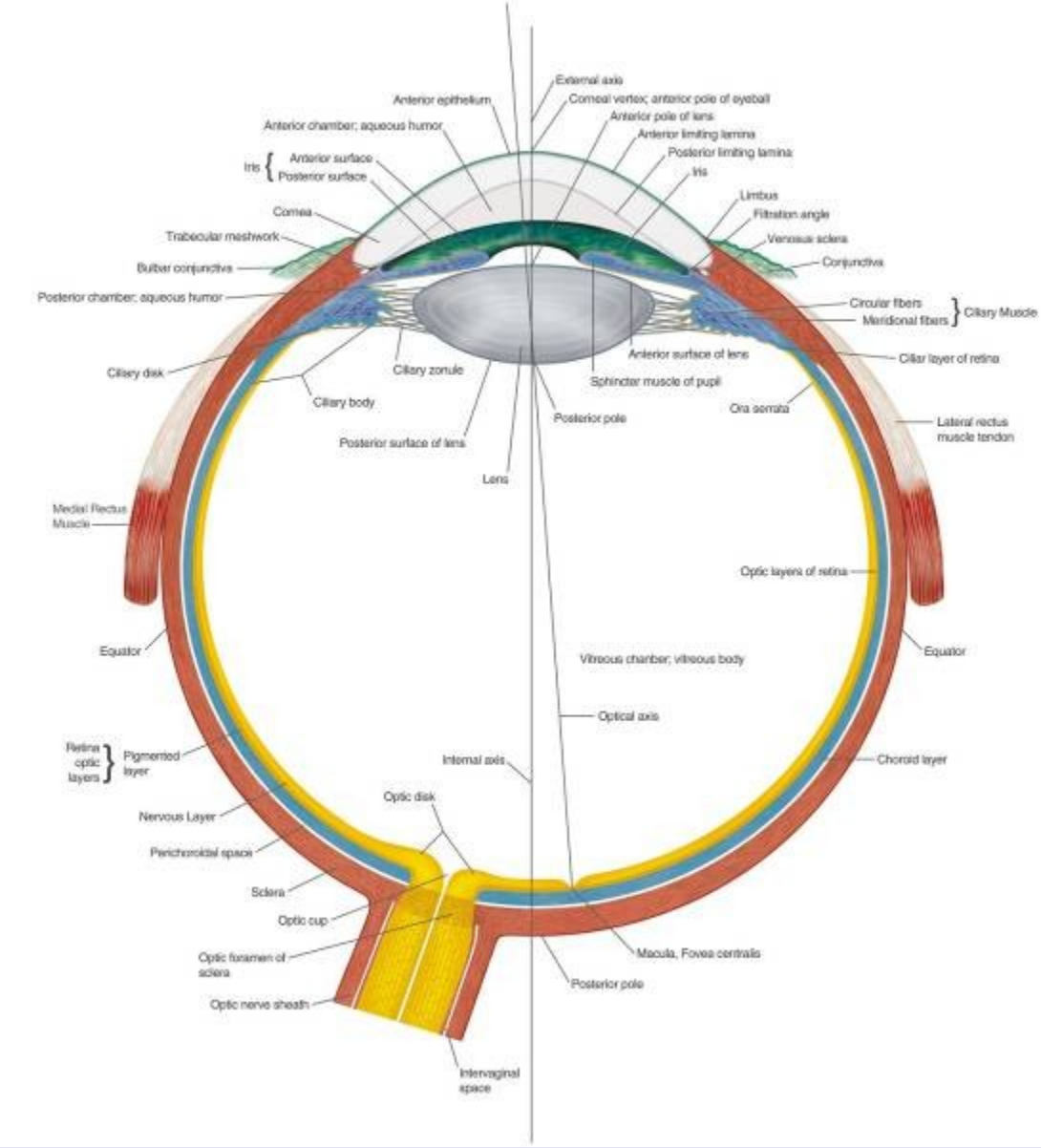


- Sclera can be regarded as cup-like expansion of dural sheath of the optic nerve.
- Choroid is an expansion of the arachnoid and pia matter
- Retina could be regarded as expansion of the brain substance.





- The eyeball is composed of **three concentric coats**.
- **External (Fibrous) coat:** that comprises of, sclera and cornea.
- **Middle (Vascular) coat:** also called uvea or uveal tract that consists of choroid, ciliary body and iris.
- **Internal (nervous) coat:** i.e. retina.



FIBROUS COAT

- **SCLERA**

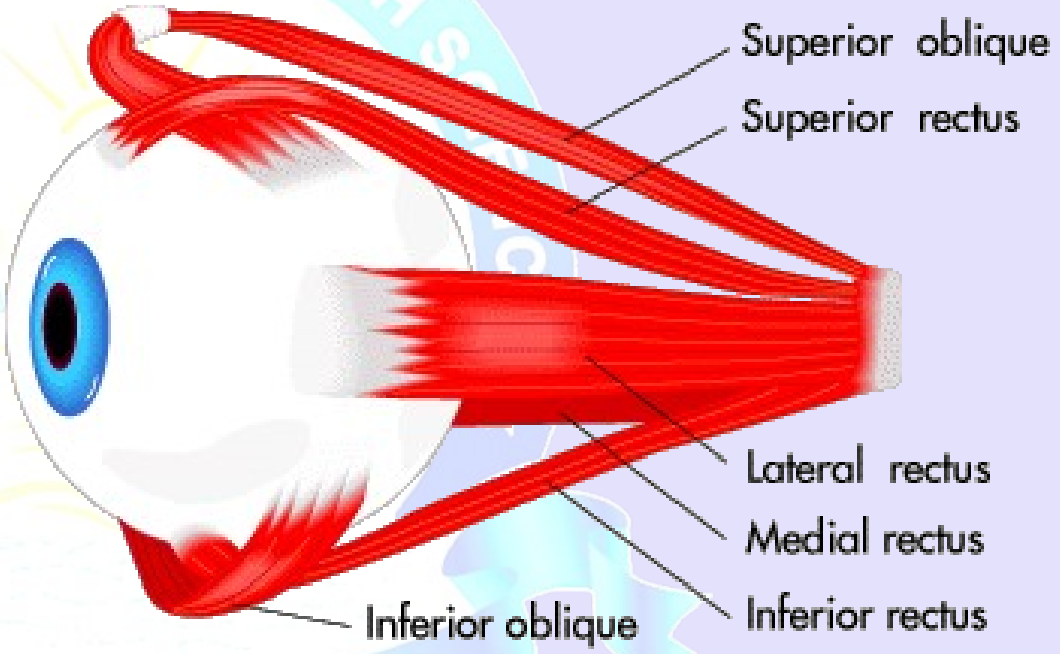
- **Posterior 5/6** of the external or fibrous coat and is **opaque** and white in colour

- Thickness about **0.5 mm**

- Composed of **dense collagen fibrous** interspersed with **elastic fibers**

- Help to maintain the size and shape of the eye ball

- It is **thinnest at equator** where it is pierced by recti thickest at back and **weakest at enterace point of optic nerve.**



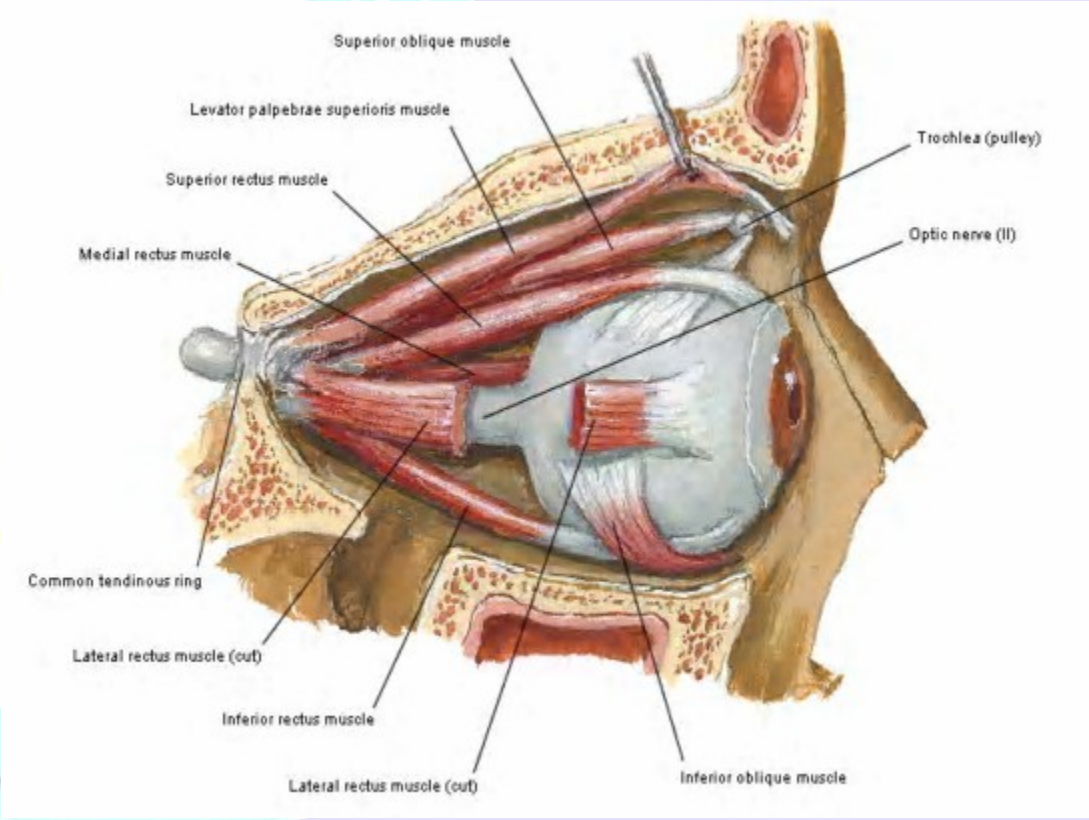
FIBROUS COAT

- **SCLERA (Cont.)**

- **Crossing fibers of optic nerve** give it sieve like appearance. This point bulges posteriorly in glaucoma causing **cupping of disc**

- Receives insertions of extra ocular muscles and is also pierce by vessels but itself **is avascular** except at circumcorneal junction

- Just behind **corneoscleral junction is cannal of schlemm** lined by endothelium drains aqueous humour in to anterior scleral veins



FIBROUS COAT

- **CORNEA**

- It is continuous with sclera with difference that its lamina are transparent instead opaque;

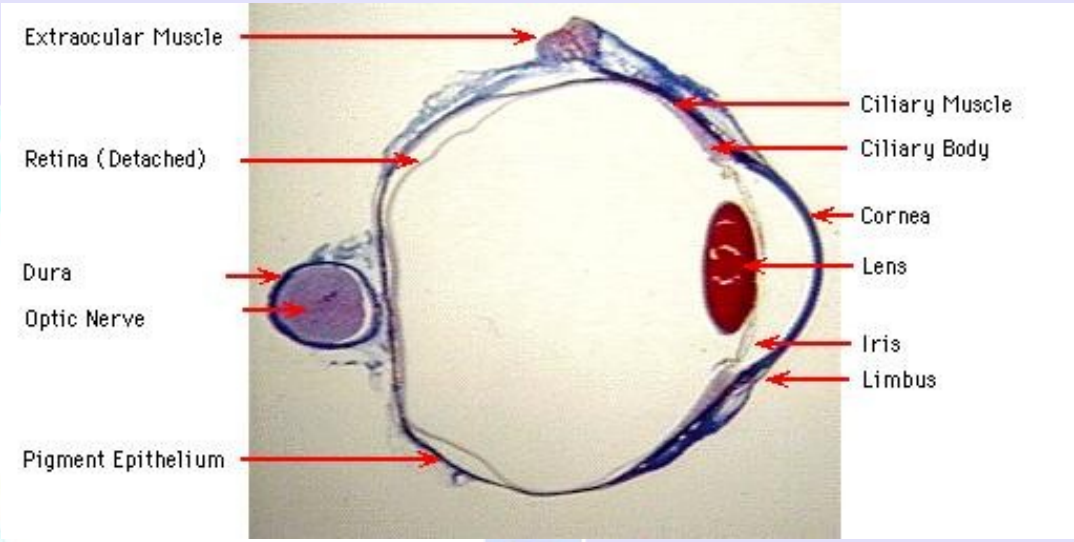
- It bulges forwards from sclera at corneoscleral junction.

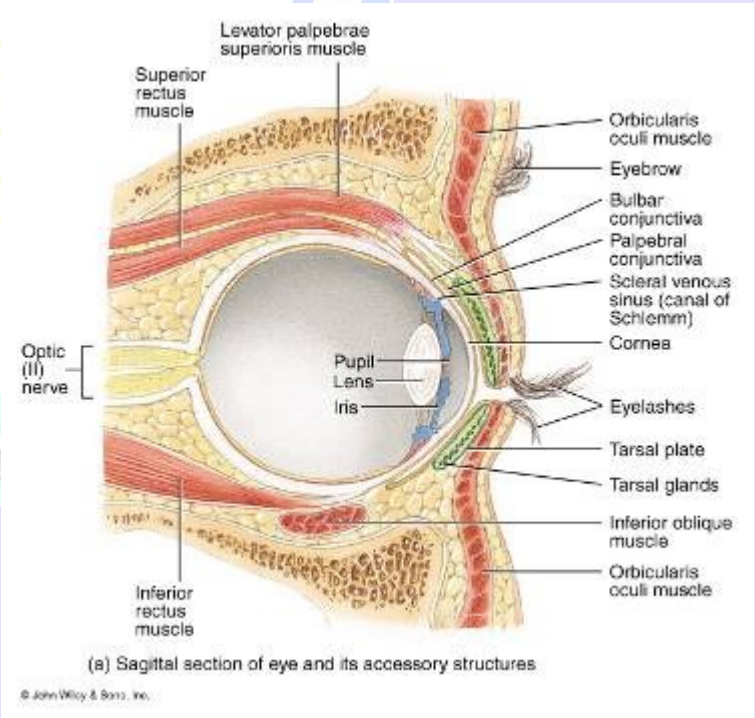
- Cornea is supplied by long and short ciliary nerves

- Corneal reflex pathway is through trigeminal nerve ganglion to main sensory nucleus from where to reticular formation to reach both facial nerve nuclei so stimulate orbicularis oculi to close both lids.

FIBROUS COAT

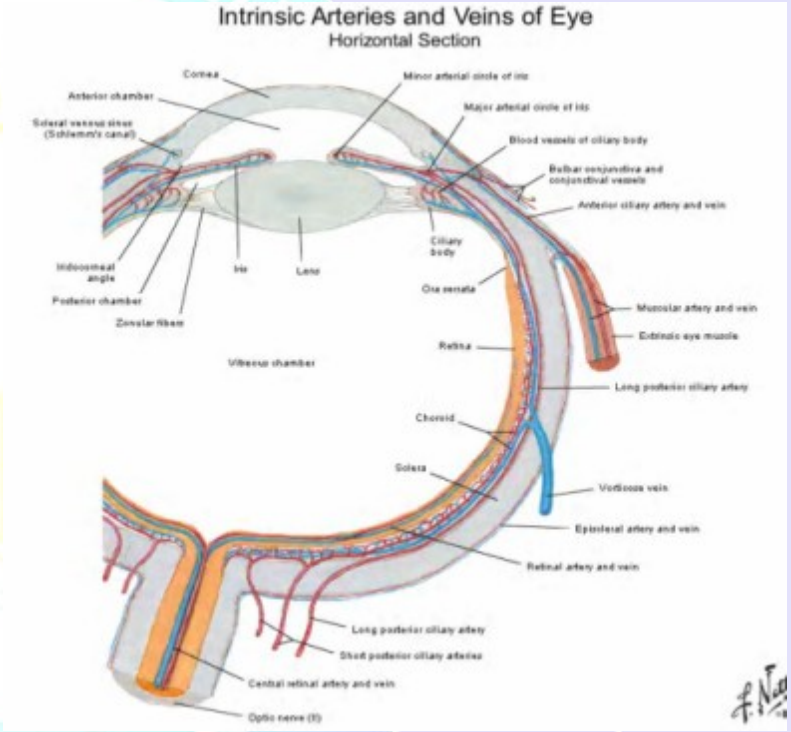
- **CORNEA (Cont.)**
- Anterior 1/6th of the fibrous coat.
- Colourless and transparent.
- Thickness about 0.8 mm in the center and 1.0 mm at the periphery.
- Cornea is an avascular structure





VASCULAR COAT

- Intermediate coat also known as uveal tract
- Consists of choroid, the ciliary body and iris

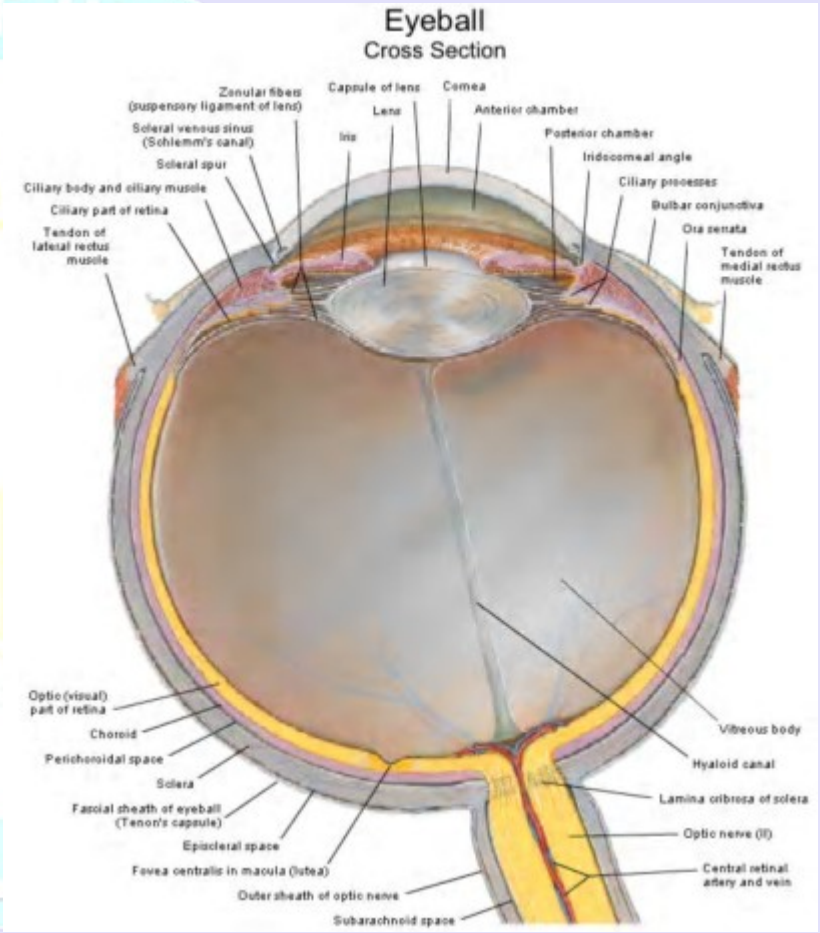


VASCULAR COAT

- **CHOROID**

- Is thin pigmented layer lining inner surface of sclera. anteriorly merges with ciliary body.

- Posteriorly perorated by optic nerve its inner surface where it is firmly attached with it, and to pigment layer of retina and choroid capillaries provides nutrition for rods and cones of retina



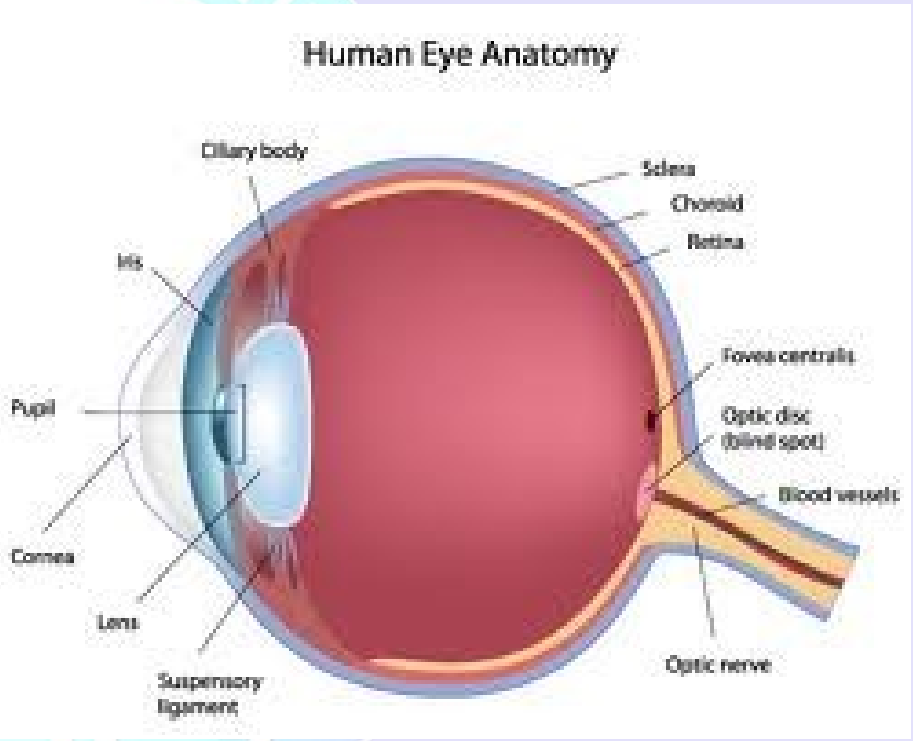
VASCULAR COAT

- **CHOROID (Cont.)**

- Veins collect into four or five large vena vorticosae which pass through equator

- Forms the posterior part of uvea or uveal tract

- Externally it is separated from the sclera by a potential space called “perichoroidal space.”



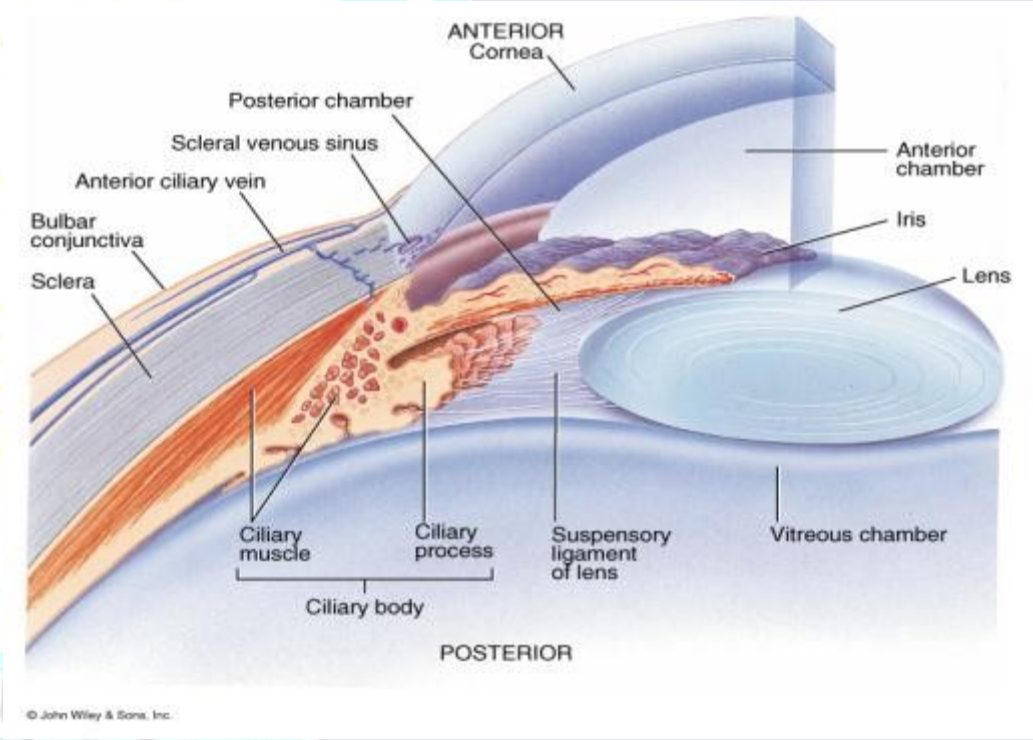
VASCULAR COAT

- **CILIARY BODY**

- Continuous with choroid behind and iris anteriorly

- It lies as flat ring applied to the inner surface of sclera
- Thicker in front and thinner behind and appears triangular

- Periphery of the iris is attached half way along the short anterior base of the triangle

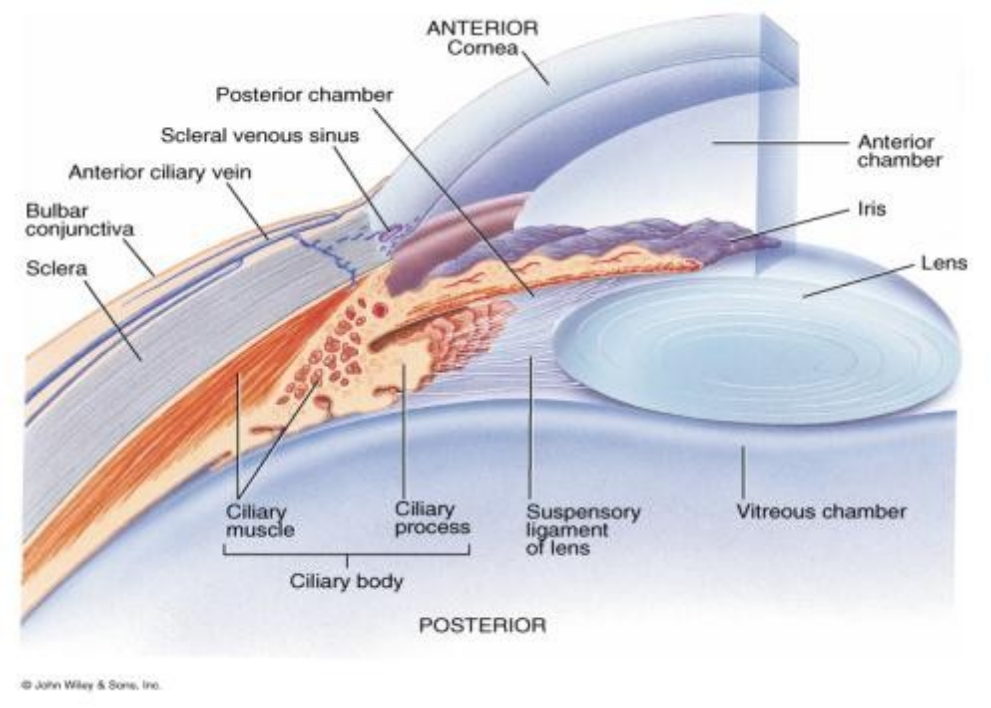


VASCULAR COAT

- **CILIARY BODY(Cont.)**

- Scleral surface contains scleral muscle

- Vitreous surface of ciliary body is lined by two layers of epithelium outer is pigmented and inner is non pigmented they represent pigment layer and nervous part of the retina this surface is continuous with choroid at ora serrata further forward this surface is projected into 70-80 small ciliary processes which lie in reciprocal grooves on the anterior surface of the vitreous body



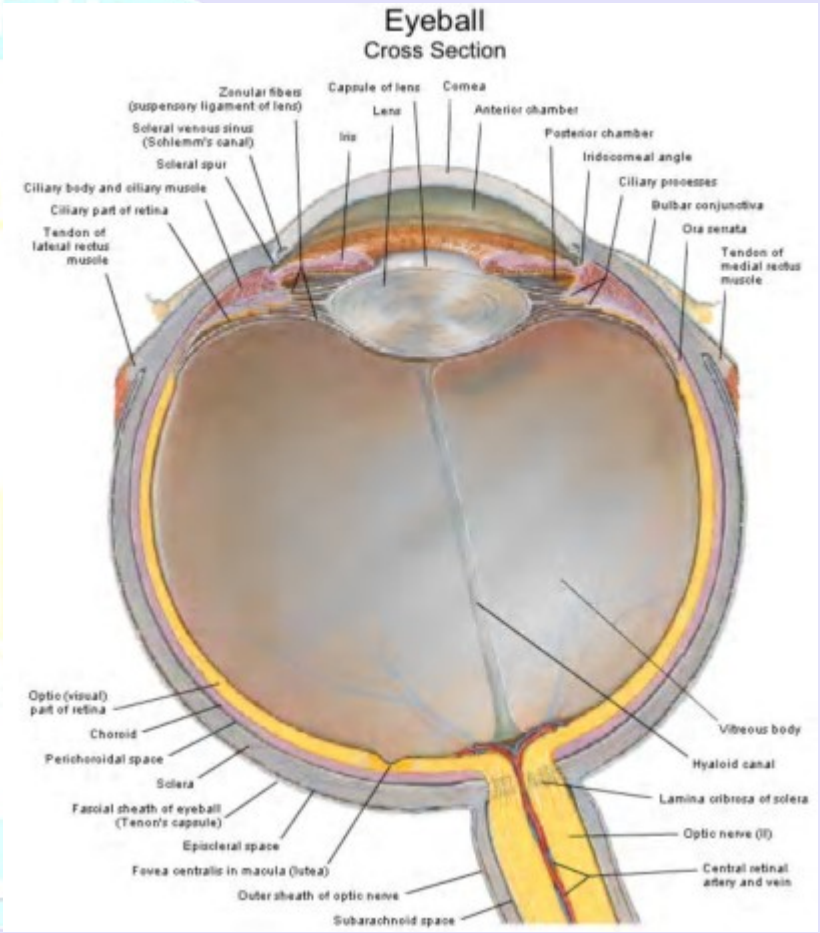
VASCULAR COAT

- **CILIARY MUSCLE**

- Smooth muscle functions to focus the length for near vision

- Outermost fibers are longitudinal and pass back into the stroma of choroid innermost are circular and run circumferentially near the periphery of lens between two are radial fibers which radiate from the scleral spur
- Contraction of ciliary muscle relaxes the suspensory ligaments allowing the lens to bulge and focus for near vision

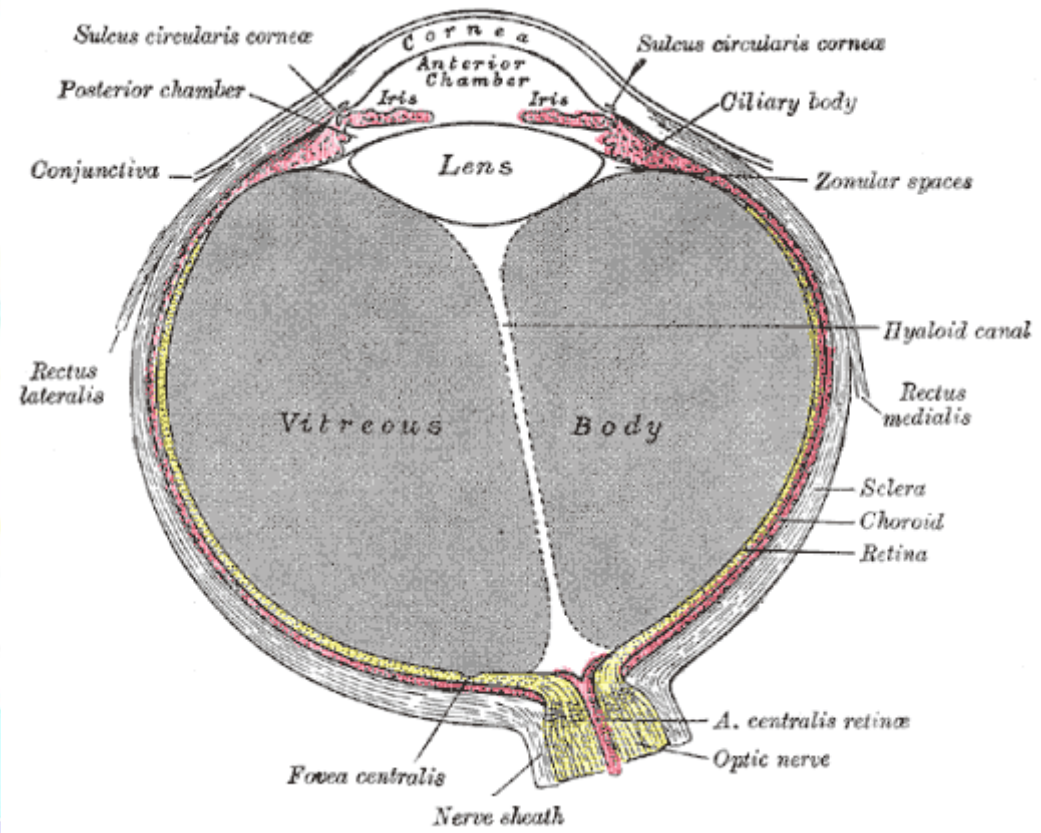
- Supplied through Edinger - westphal nucleus of oculomotor nerve



VASCULAR COAT

- **IRIS**

- Attached to middle of anterior surface of ciliary body
- Iris is pushed slightly forward by narrow convexity of the lens
- Iris is perforate in the central by pupil
- Pupil controls the light entrance
- Iris has made up of vascular tissue and with melanocytes
- Colour of the iris is determined by the amount of pigment in the iris
- Iris lacking pigment is blue



VASCULAR COAT

- **SPHICTER PUPILLAE**

- Is circular band of smooth muscles lying in the stroma of the iris at the margin of the pupil

- Supplied through Edinger westphal nucleus

- Dilator pupillae is thin sheet of radial fibers of smooth muscle at the back of stroma of the iris extending from ciliary body to the sphincter pupillae . Supplied by cervical sympathetic T1

- Stroma of iris and dilator muscle are derived from neural crest origin

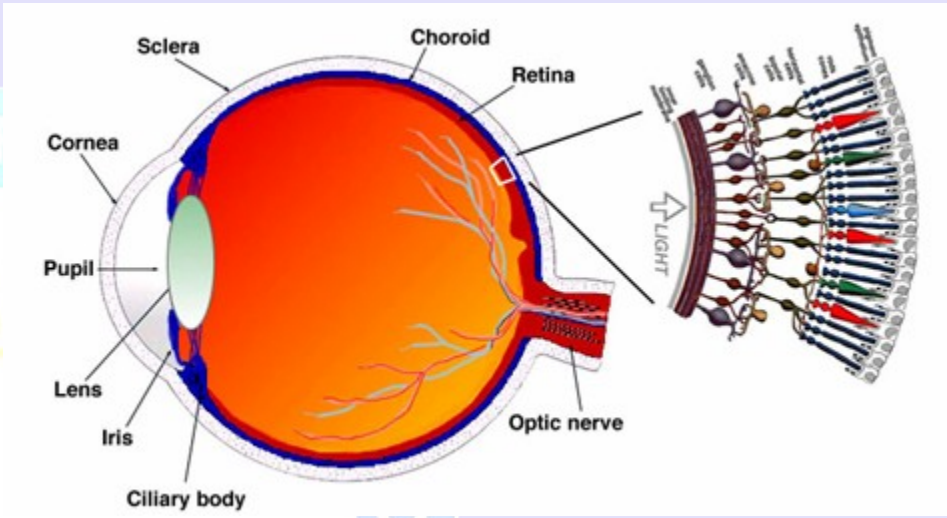
NERVOUS COAT

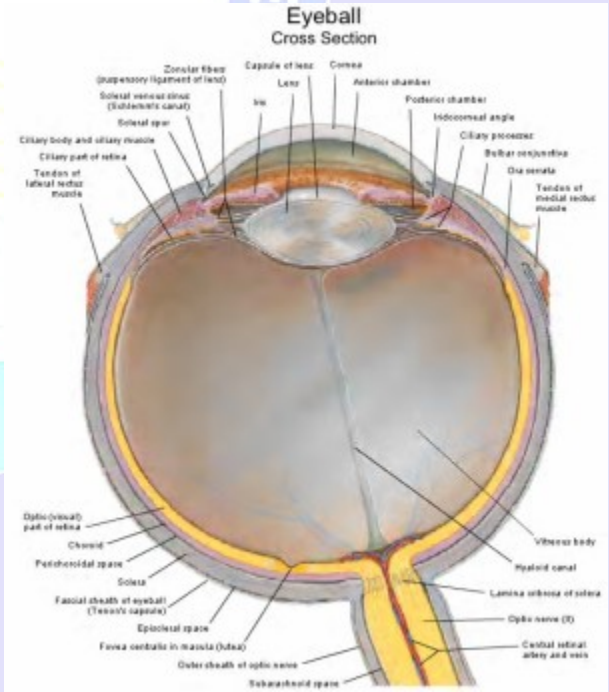
- **RETINA**

- Outer surface is in contact with choroid and inner with vitreous body

- Light sensitive area ends abruptly halfway between equator and corneoscleral junction at ora serrata

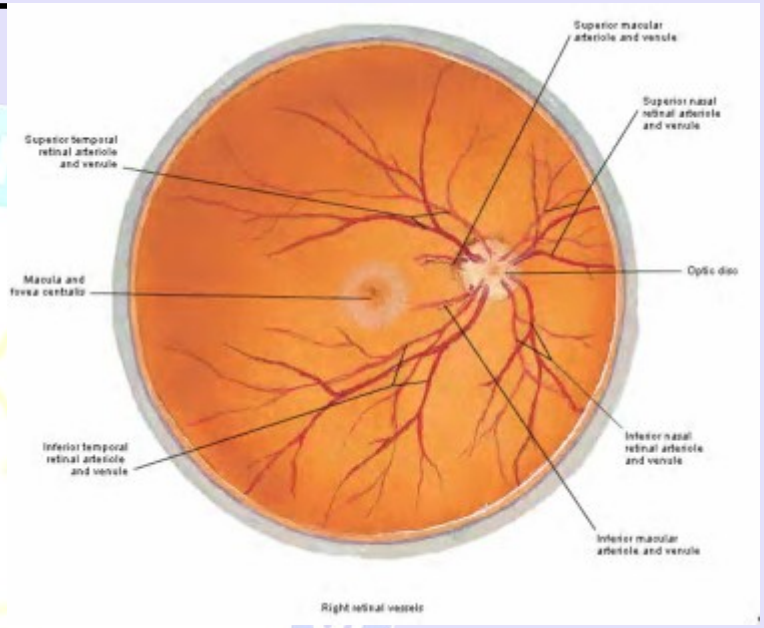
- At entrance of optic nerve is a circular area pale area, 1.5 mm in diameter this is optic disc





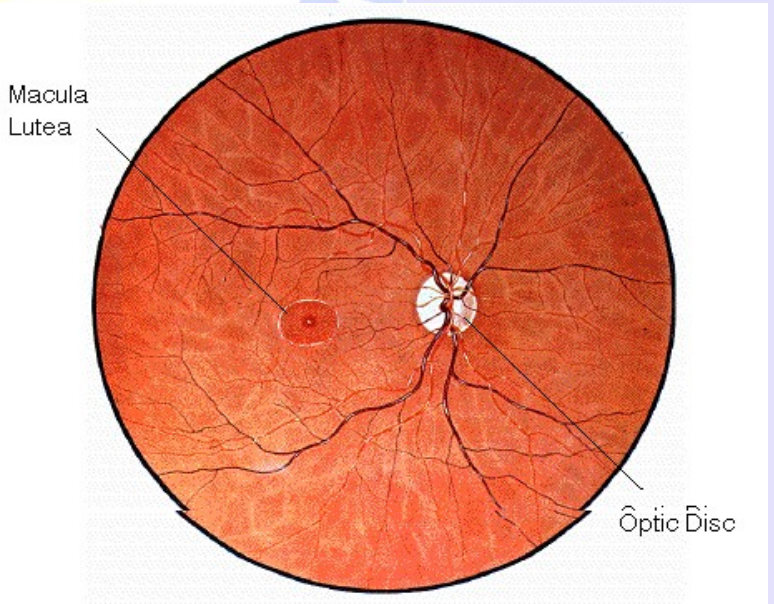
NERVOUS COAT

- **RETINA (Cont.)**
- Optic disc is excavated to a a variable degree producing physiological cup having no rods and cones there fore it is blind area



- Disc and surrounding area is fundus of eye

- 3 mm lateral to optic disc is a shallow depression, yellow called macula lutea
- In the center of macula is shallow pit the fovea centralis of comparable size to disc (thinnest part of retina with no blood vessels and no rods here)



NERVOUS COAT

- **RETINA (Cont.)**

- Light receptors are of two kinds rods which are sensitive dim light scotopic vision

- Cones have register colour lighter threshold

- Central artery of retina passes through the lamina cribriosa with in optic nerve and in optic disc and divides into upper and lower branch its branches are end arteries

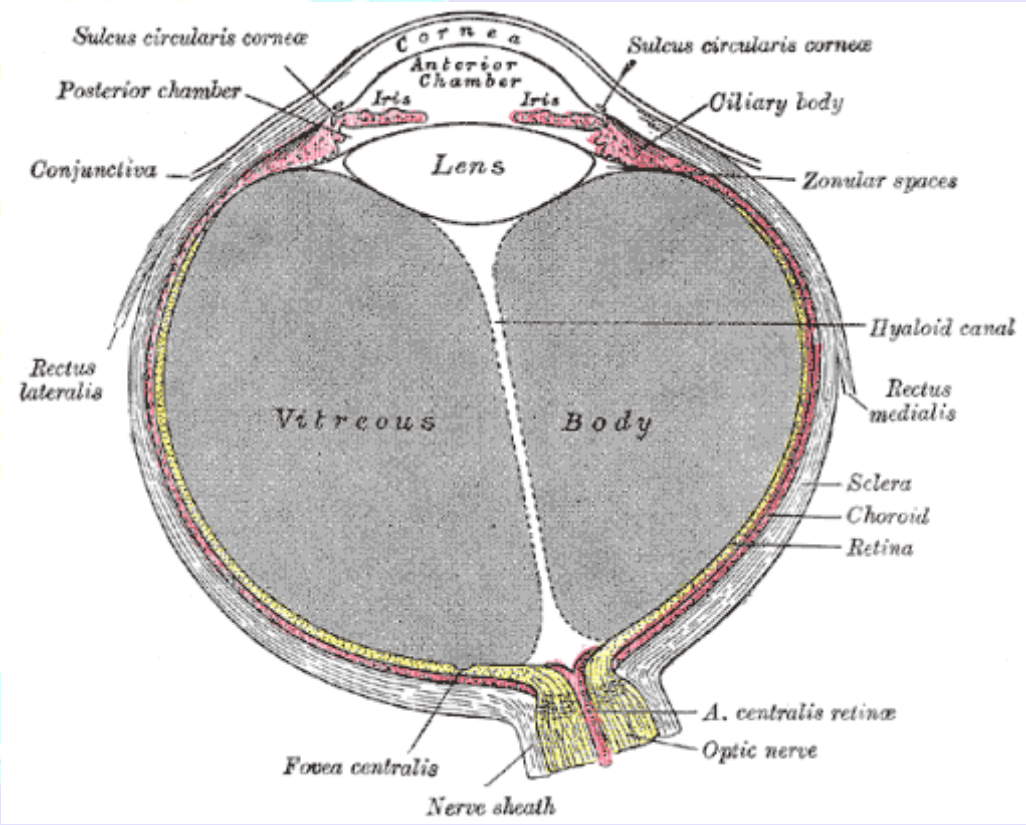
- Retinal vein run with branches of the arteries pass via optic disc and emerges from optic nerve and its coverings to join superior ophthalmic vein

AQUEOUS HUMOUR

- Clear fluid lies between back of cornea and front of the lens
- Space is divided by iris into anterior chamber and posterior chamber
- Chamber communicate with each other through pupil.
- Anterior chamber is 3mm deep centrally

AQUEOUS HUMOUR (Cont.)

- At margin of anterior chamber it is filtered through trabecular tissue into canal of schlemm
- Obliteration of angle can block its drainage causing glaucoma
- It is source of nutrition and metabolic to the cornea
- Posterior chamber is bounded infront by the iris behind by the lens and its suspensory ligament

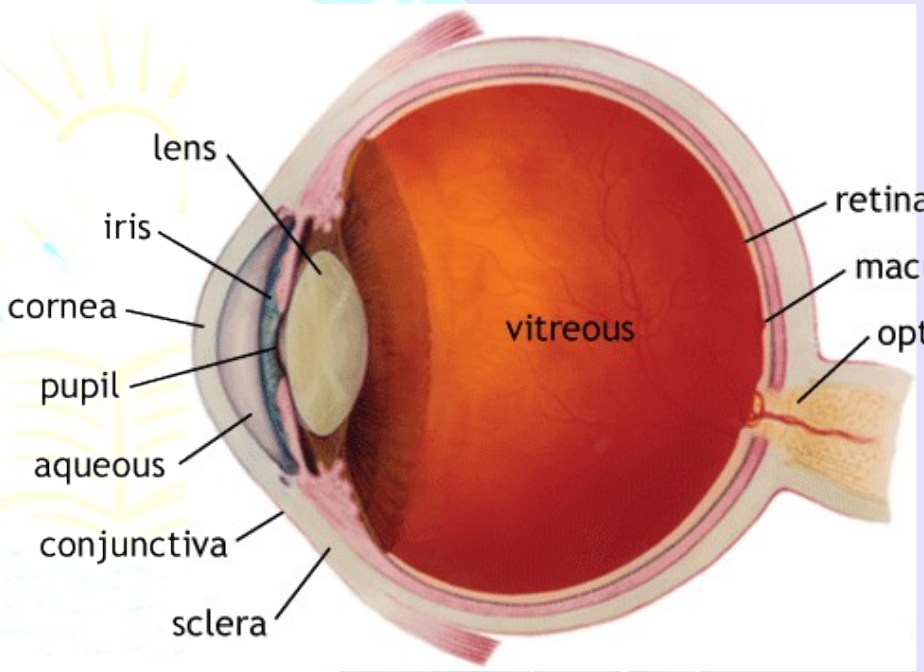


VITREOUS BODY

- Occupies the region of the eye behind the lens.

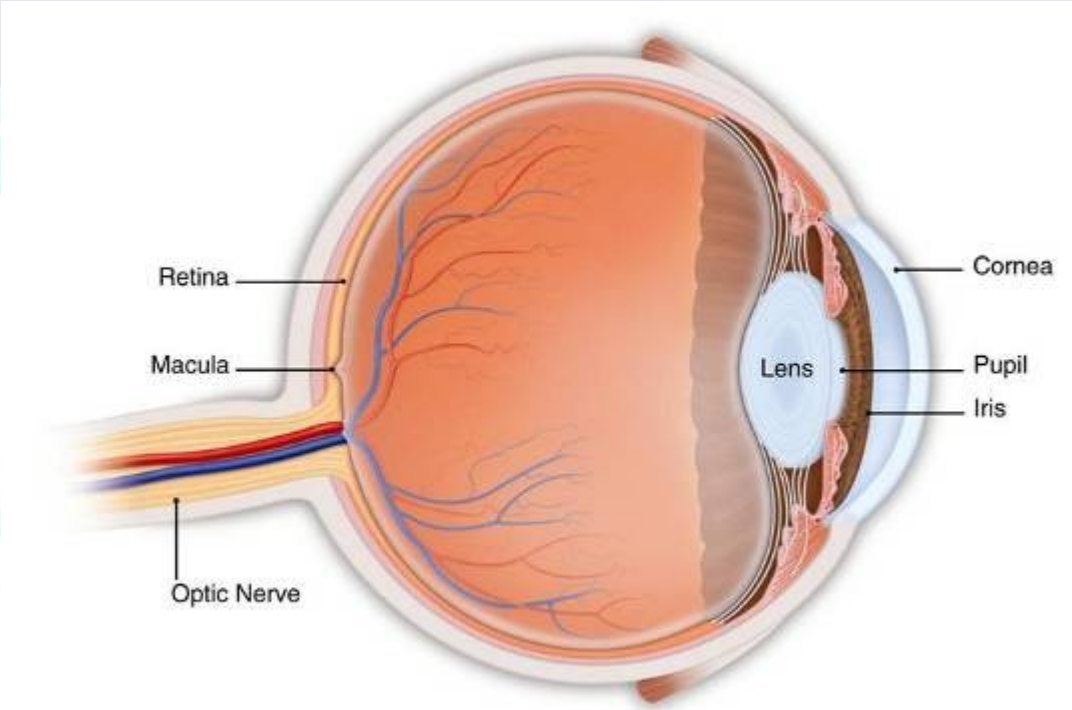
- A transparent gel that consists of water (about 99%),a small amount of collagen heavily hydrated with hyaluronic acid molecules and very few cells (hyalocytes).

- Hyalocytes synthesize collagen and hyaluronic acid.



LENS

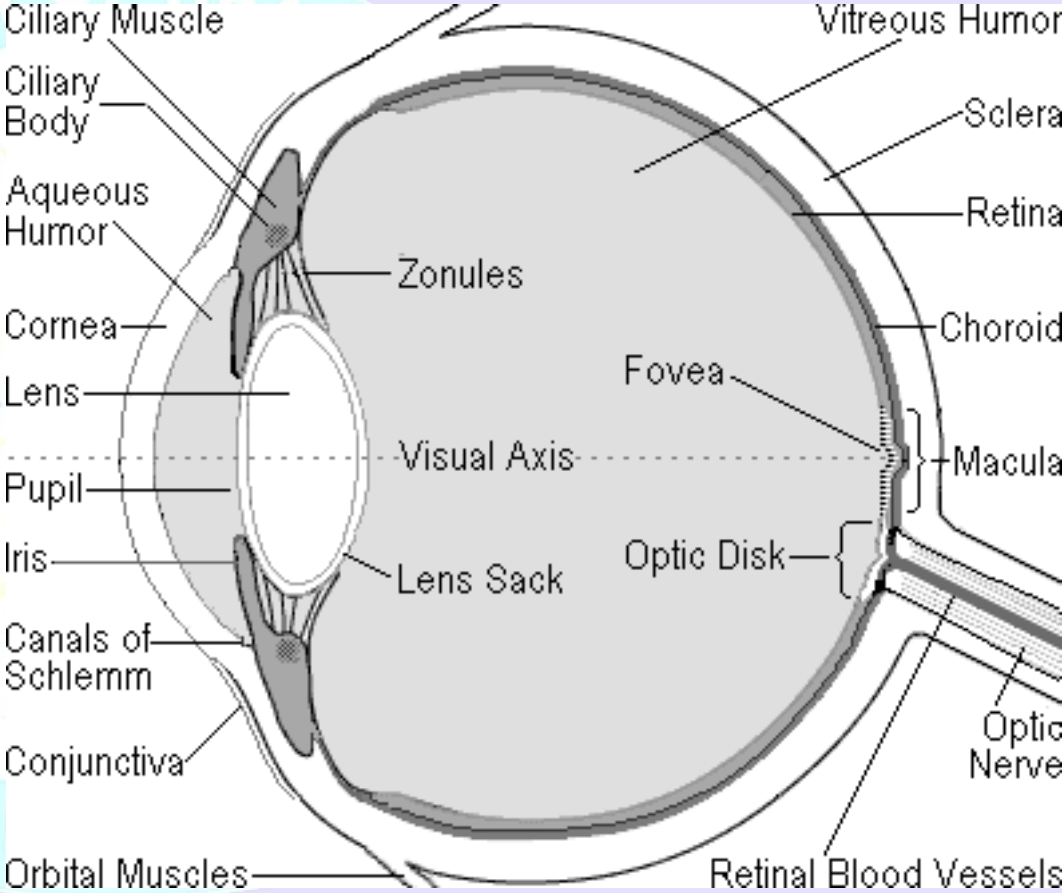
- Transparent biconvex body enclosed by elastic capsule
- 10 mm in diameter 4mm thick
- Posterior part resting on vitreous is more convex
- Anterior surface is in contact with iris





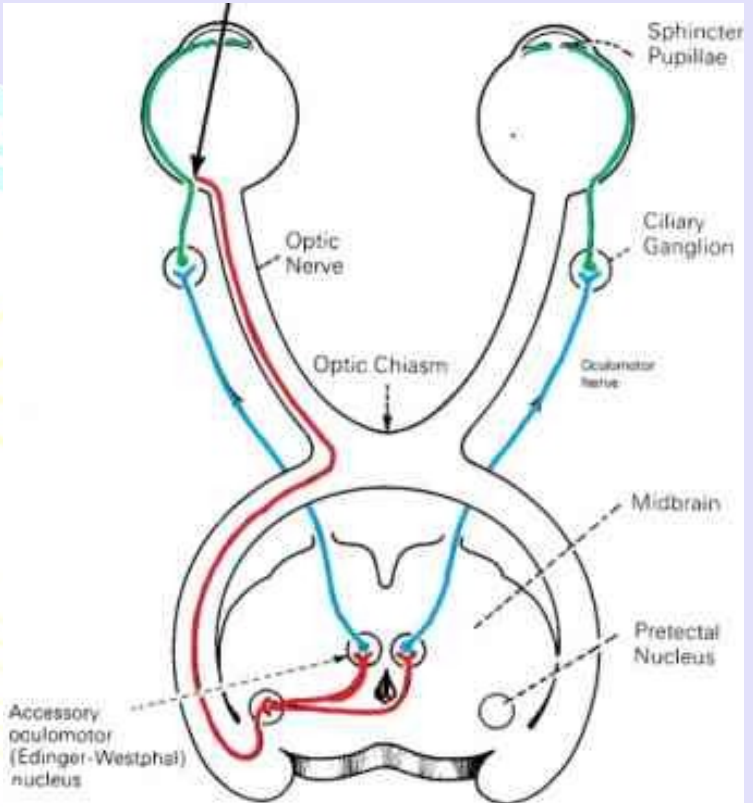
SUSPENSORY LIGAMENT

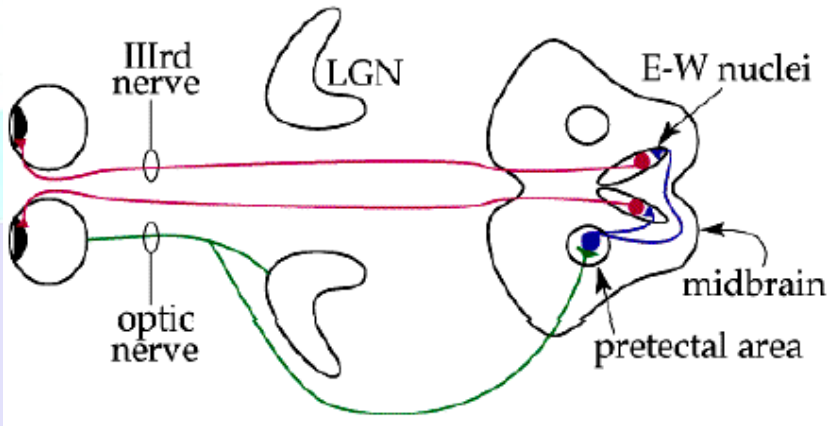
- Suspensory ligament of lens or zonule is a series of delicate fibrils attached to the ciliary process and through the furrows between them further back on the ciliary body
- The fibers pass centrally to attach themselves to the lens mostly infront of but few behind the circumference
- In the rest position they hold the lens flattened under tension when relaxed by the contraction of ciliary muscle. The elasticity of the causes its anterior surface to bulge so thickening it



PUPILLARY LIGHT REFLEX

- Direct pupillary light reflex pupil constricts pupil when light is shone into eye
- The other eye also constricts, it is called consensual light reflex.
- **Pathway**: Some fibers of the optic tract pass lateral geniculate body and reach the midbrain at the level of superior colliculus and enter in to pretectal nucleus, from there fibers enter in to Edinger Westphal nucleus (bilaterally). From it fibers pass to ciliary ganglion via oculomotor nerve. Then through short ciliary nerve fiber reach to sphincter pupillae.
- **Consensual light reflex** is due to Edinger Westphal nucleus receiving fibers bilaterally





ACCOMMODATION CONVERGENCE REFLEX

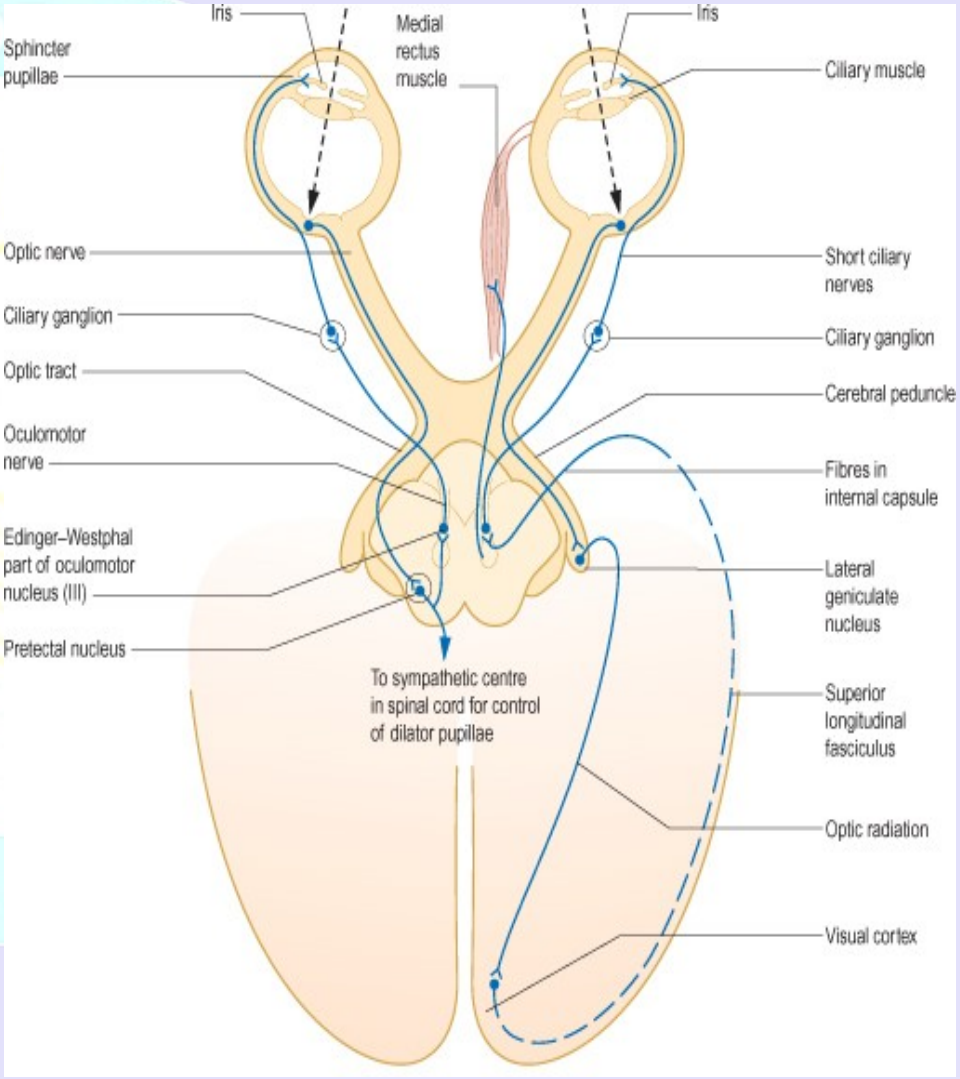
- Eye automatically focuses and converges for near vision, and this change in lens curvature is accompanied by pupillary constriction to sharpen focus. It has three components

1. Accommodation,
2. Convergence and
3. Pupillary constriction constituting accommodation-convergence reflex or near reflex.

- From the visual cortex an association bundle reaches the frontal eye field of middle frontal gyrus.

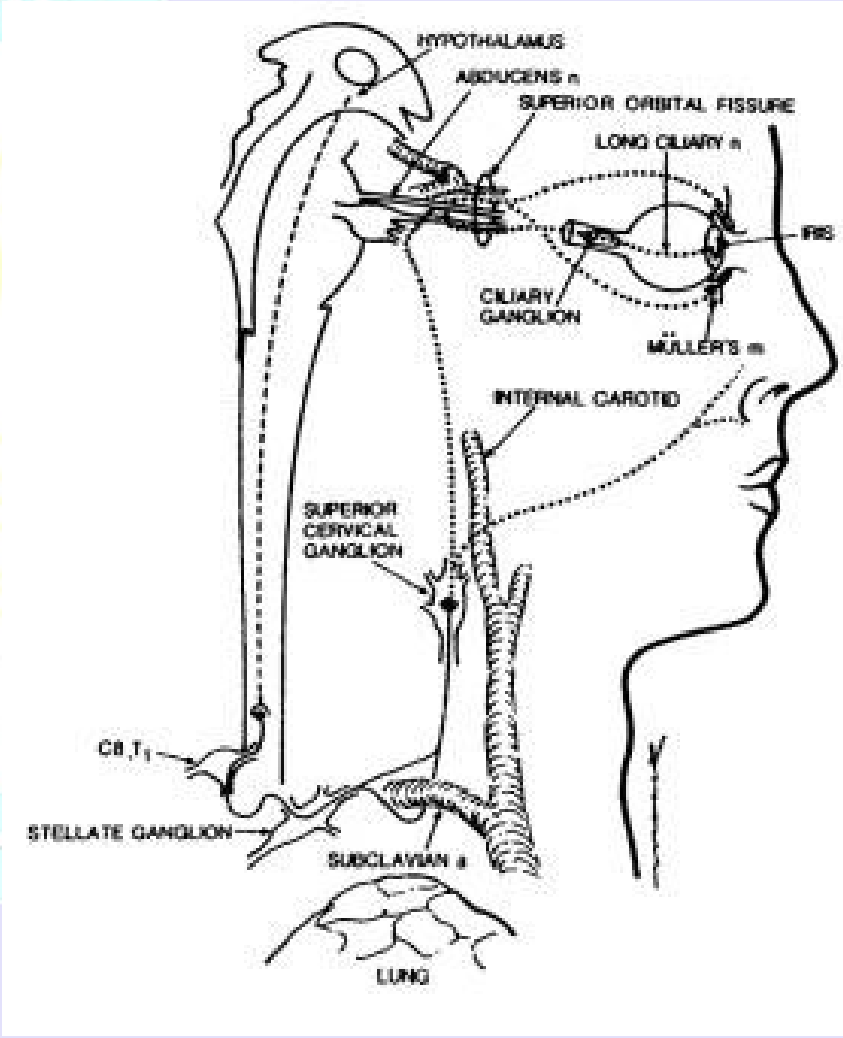
- Cell bodies in this cortex send their efferent fibers via anterior limb of internal capsule to the oculomotor nucleus including Edinger Westphal part from where by ciliary ganglion.

- This relay activates the ciliary muscle and the sphincter pupillae. Contraction of ciliary muscle releases tension on suspensory ligament of lens so allowing lens by its own elasticity to thicken and focus for near vision. Accompanying the accommodation changes the medial rectus muscle of each eye contracts to provide the necessary convergence for near vision



DILATOR PUPILLAE INNERVATION

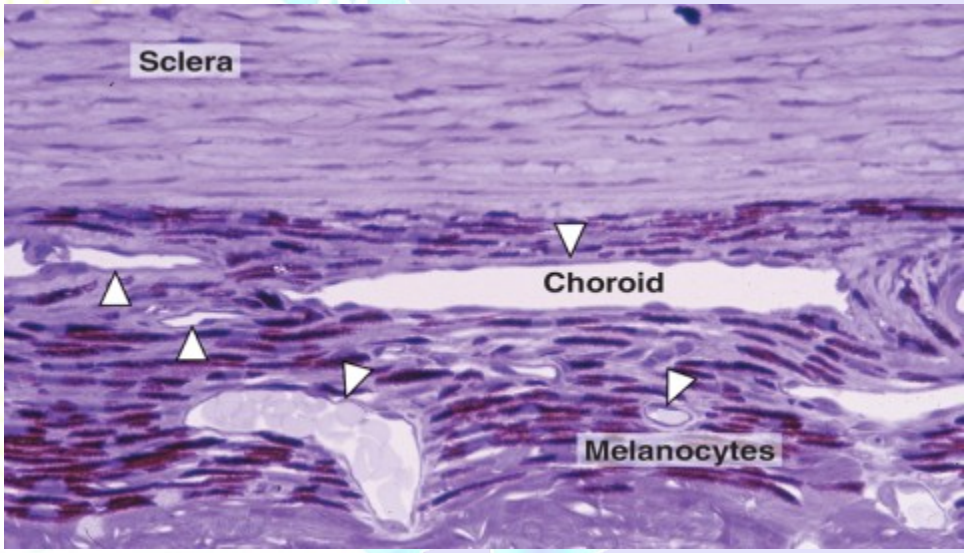
- It is through the sympathetic path which is very long.
- From cells in hypothalamus fibers run down through the brain stem and spinal cord to lateral horn cells in the T1 segment of the cord. Preganglionic fibers enter the sympathetic trunk via the white ramus communicants of T1 nerve and pass up to the superior cervical ganglion
- From there post ganglionic fibers accompany the internal carotid artery into the skull and cavernous sinus leaving the artery to join the ophthalmic nerve and become distributed to the eye by nasociliary then the long ciliary branches



HISTOLOGY OF EYE

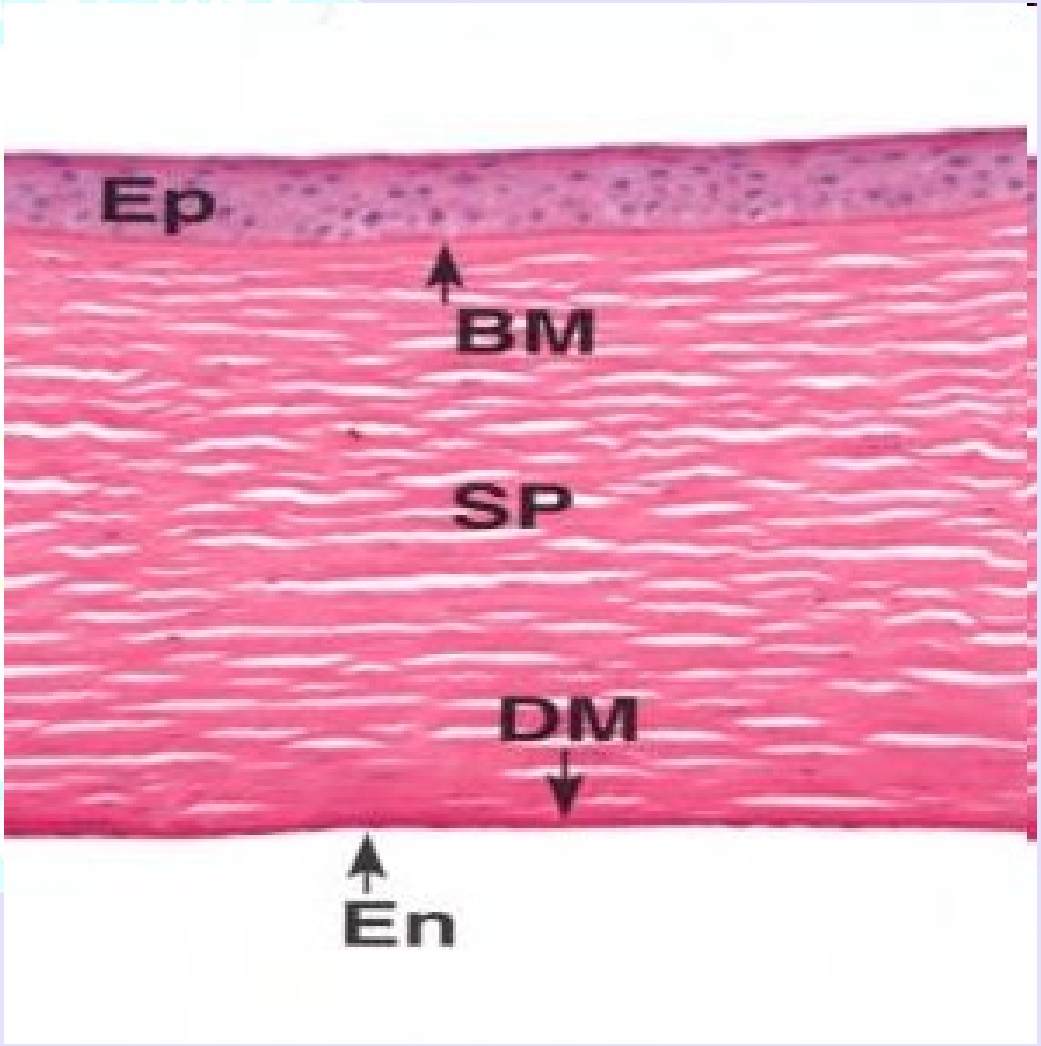
SCLERA

- From without inwards consists of 3 layers:
- **Episcleral tissue:**
- Loose collagenous and elastic fibers and large number of blood vessels.
- **Sclera proper:**
- Flat bundles of collagenous fibers lying parallel to surface, some elastic fibers, moderate amount of ground substance and few fibroblasts.
- **Lamina fusca (dark layer):**
- Thin bundles of collagen fibers numerous elastic fibers fibroblasts and large amount of melanocytes.



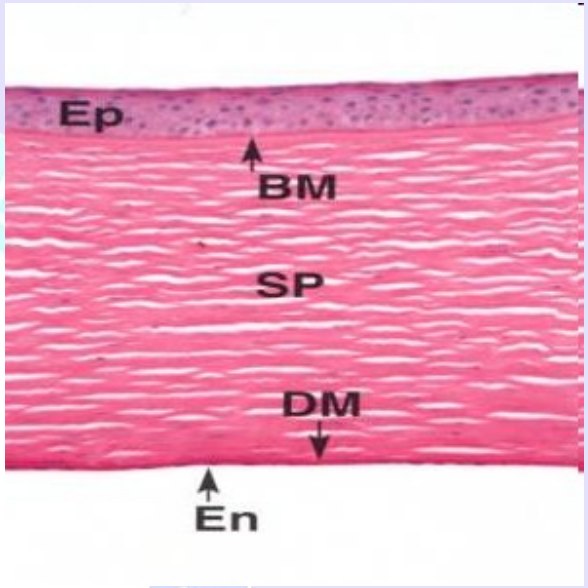
CORNEA T/S

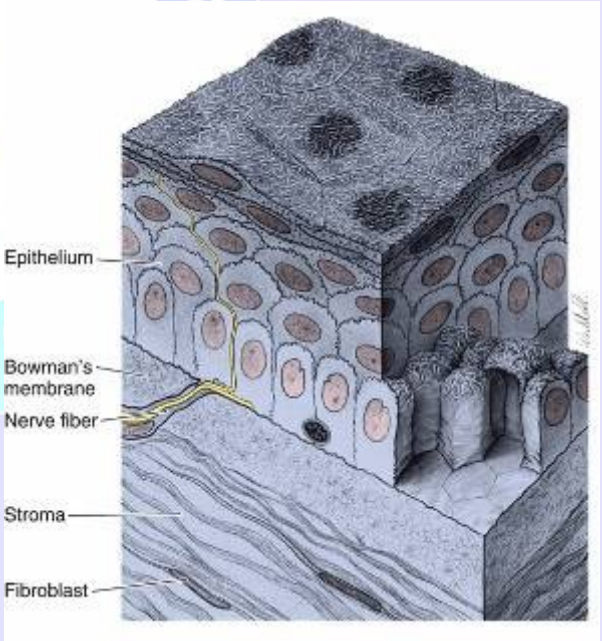
- In transverse section (from before backwards) it composed of following five layers:
- **1.** Epithelium
- **2.** Bowman’s membrane
- **3.** Substantia propria
- **4.** Descemet's membrane
- **5.** Endothelium



CORNEA

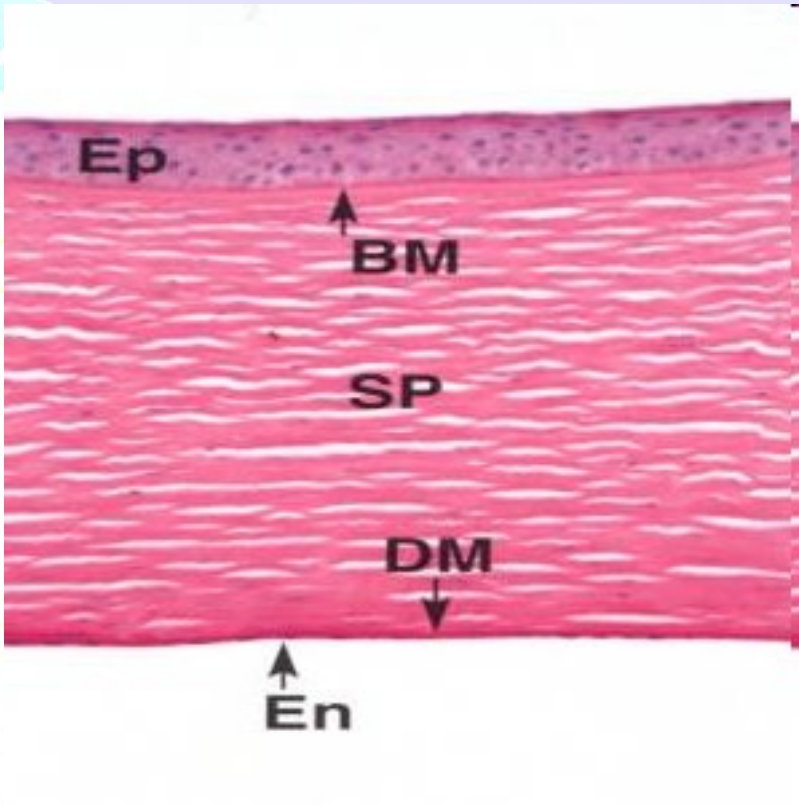
- **1. Epithelium:**
- Stratified squamous noncornified (5-7 layers) with basal layer shows mitotic figures. Highly sensitive
- **2. Bowman's Membrane:**
- Homogenous and structure, less thin layer. Consists of network of fine collagenous fibers embedded in ground substance.
- It plays role in strength and stability of cornea.





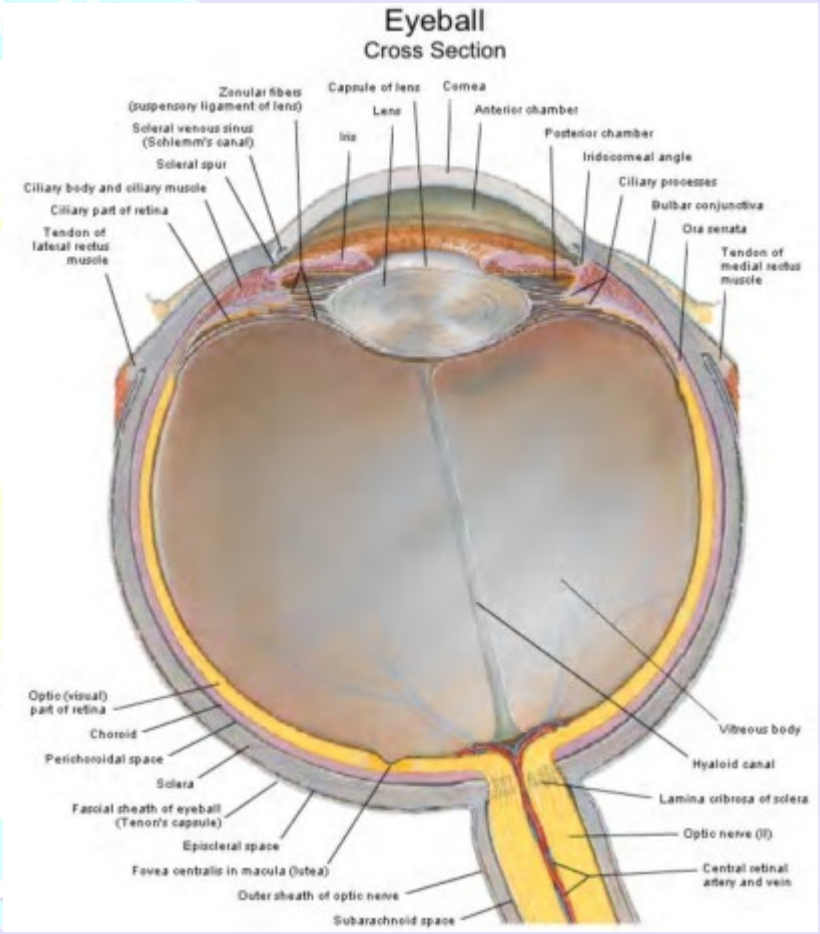
CORNEA

- **3. Substantia Propria (corneal stroma):**
- Thickest (90%).
- Composed of several bundles of collagenous fibrils bounded by glycosaminogycans
- **4. Descemet's Membrane**
- Actually it is the thick basement membrane of endothelium.
- Homogeneous and highly refractile and thin membrane.
- **5. Endothelium**
- Lines the posterior surface of cornea.
- Consists of squamous or low cuboidal cells.



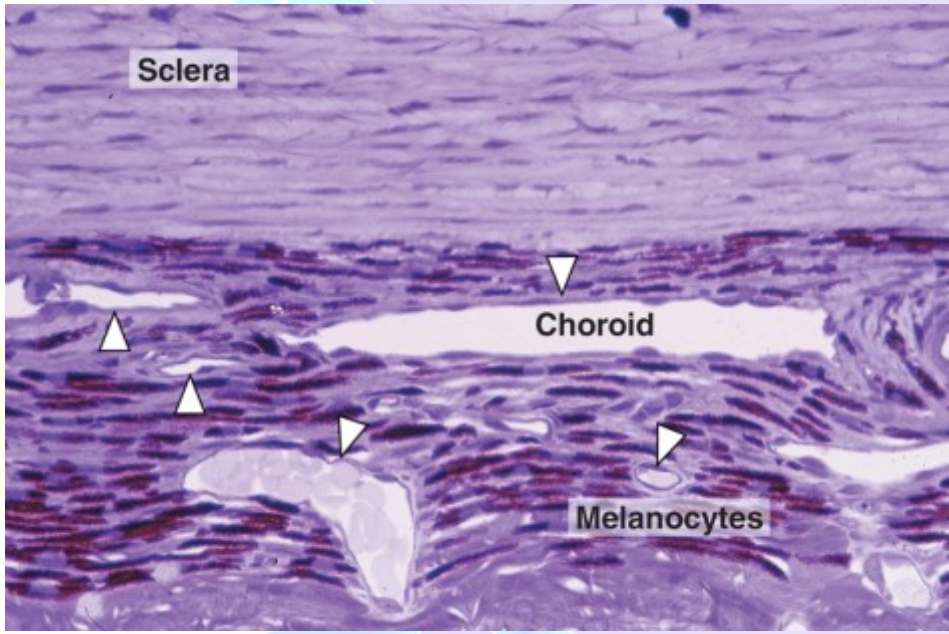
CHOROID

- Forms the posterior part of uvea or uveal tract
- Externally it is separated from the sclera by a potential space called “perichoroidal space.”
- Structurally it composed of 4 layers: (from before backwards)
- **1.** Suprachoroid
- **2.** Vessel layer
- **3.** Capillary layer
- **4.** Lamina vitrea



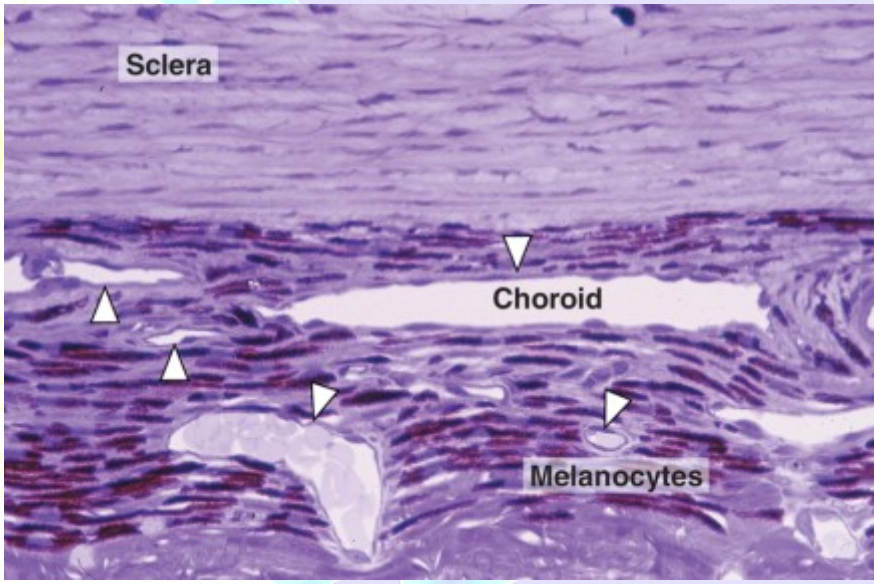
LAYERS OF CHOROID

- **1. Suprachoroid**
- A thin layer, composed of loosely arranged collagenous and elastic fibers
- **2. Vessel layer**
- Thickest layer.
- Consists of a mass of arteries and veins and a large number of melanocytes lying in loose connective tissue.
- Melanocytes gives this layer a characteristics black colour.

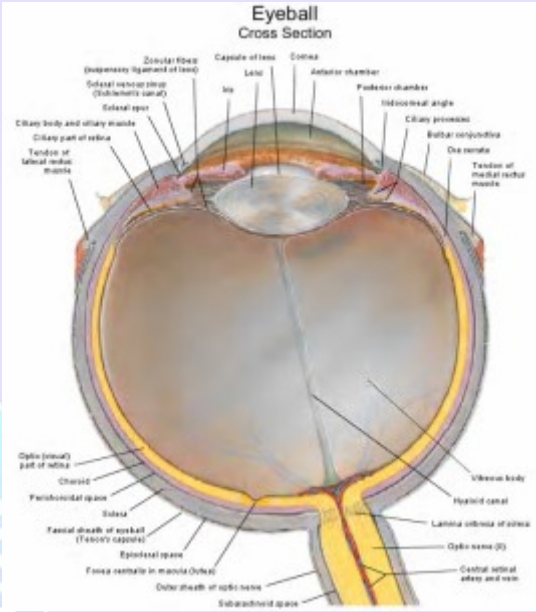


LAYERS OF CHOROID

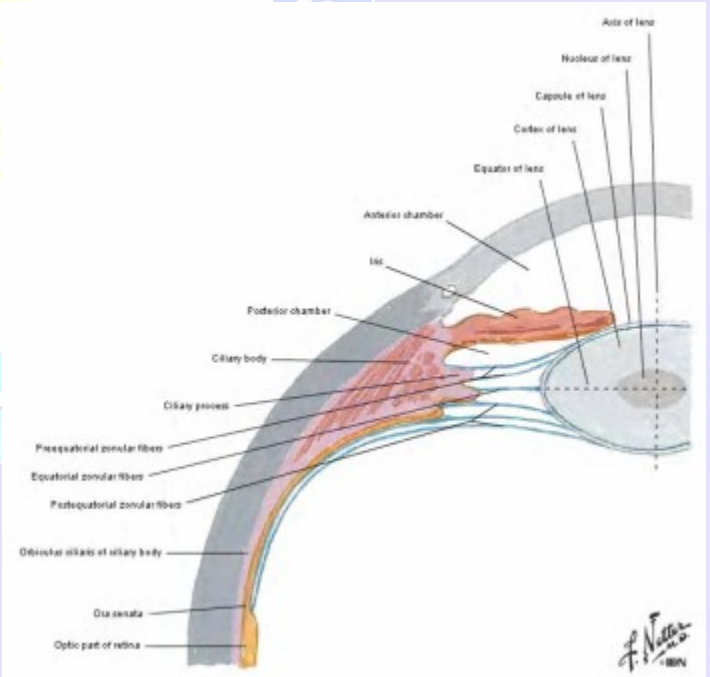
- **3. Capillary layer (choriocapillary layer)**
- Consists of a plexus of unusually large fenestrated capillaries.
- Play an important role in providing nutrition to the outer portion of Retina.
- **4. Lamina vitrea (Lamina elastica or Bruch's membrane)**
- A thin shiny, homogeneous layer that lies directly on the pigment epithelium of retina.



CILIARY BODY

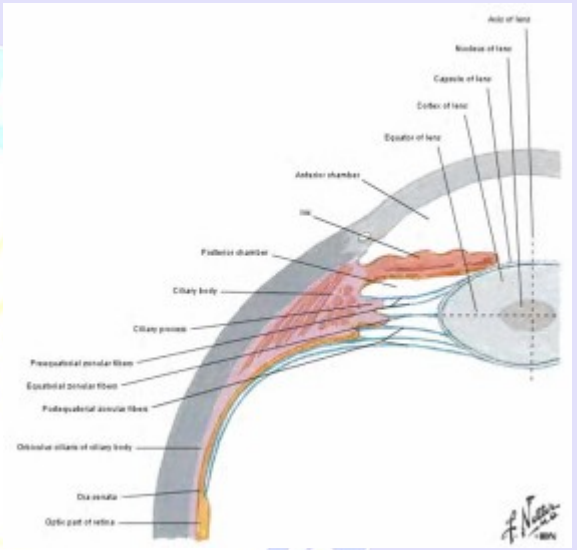


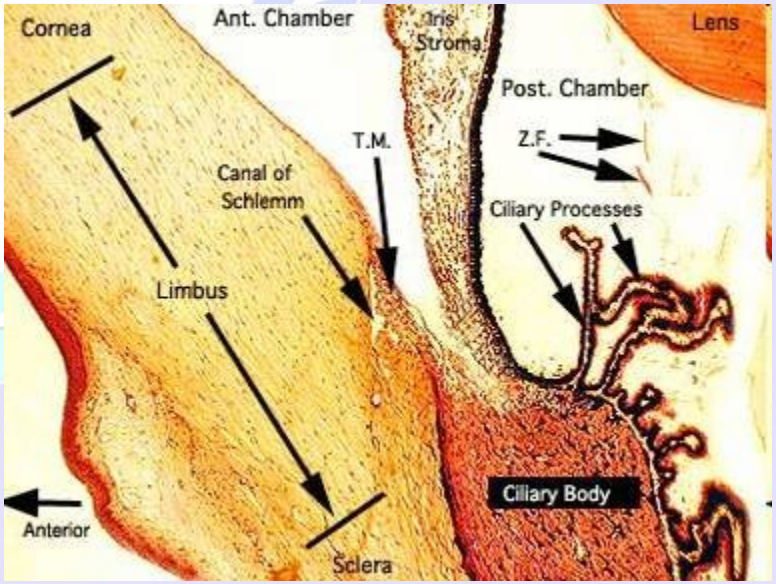
- Choroid extends anteriorly as far as the ora serrata (anterior margin of the sensory portion of retina) anterior to the ora serrata, uveal tract thickened to form ciliary body
- In L/S appears triangular in shape:
- Base faces the anterior chamber
- Outer surface blends with sclera
- Inner surface faces vitreous body



CILIARY BODY

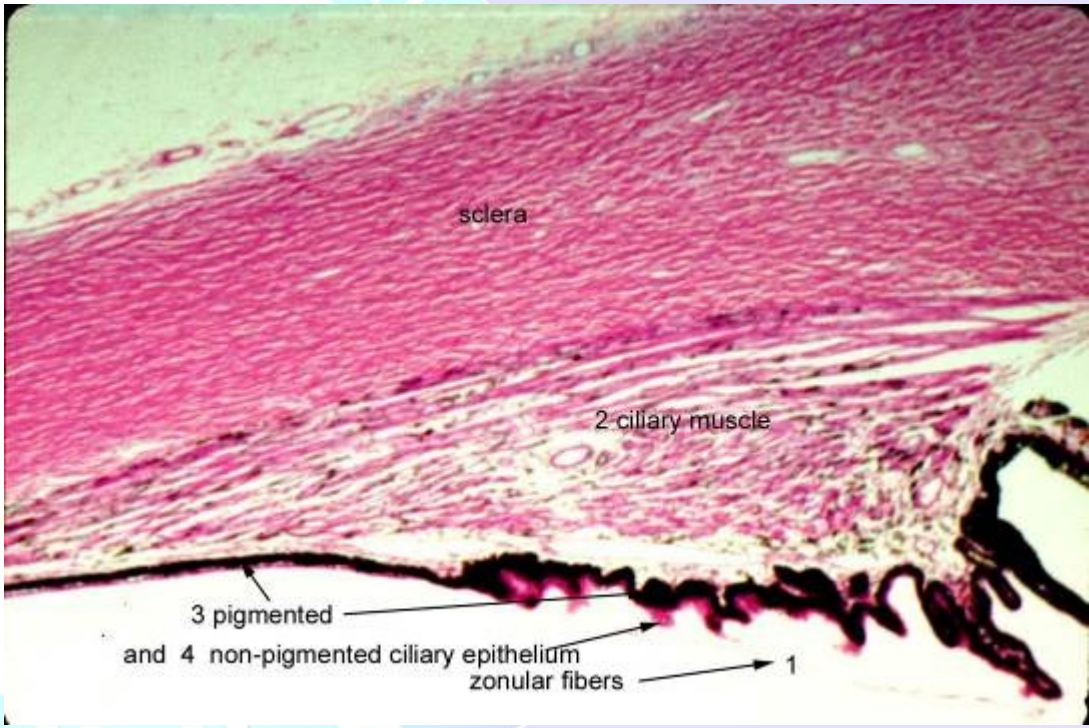
- On naked eye inner surface show 2 zones:
- Posterior 2/3rd is smooth and darkedly pigmented called “pars plana”.
- Anterior 1/3rd is pale and bears about 70 – 80 radially arranged ridges (ciliary processes) called “pars plicata.





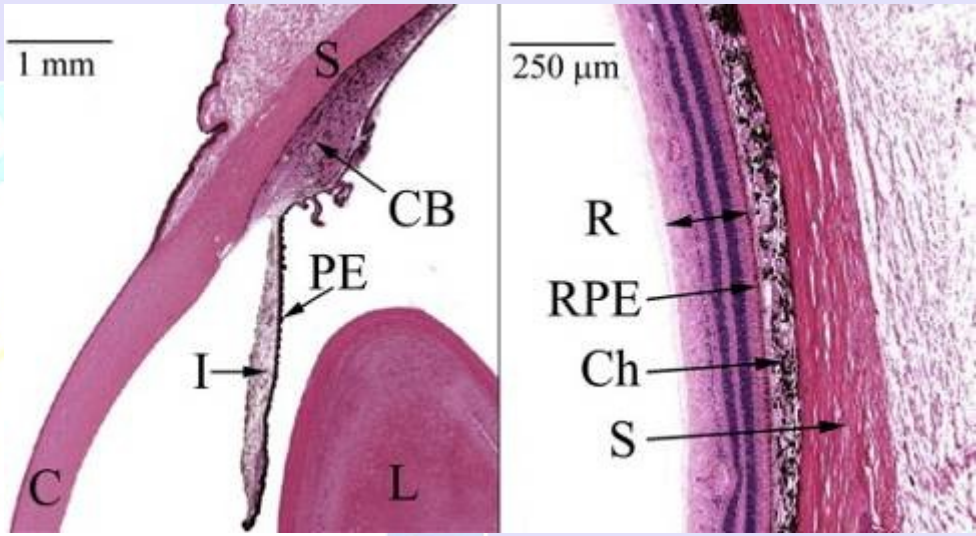
CILIARY BODY

- A section of ciliary body shows 6 layers (without inwards)
- Ciliary Muscle
- Vessel Layer
- Lamina Vitrea
- Pigmented Epithelium
- Ciliary Epithelium
- Internal Limiting Membrane

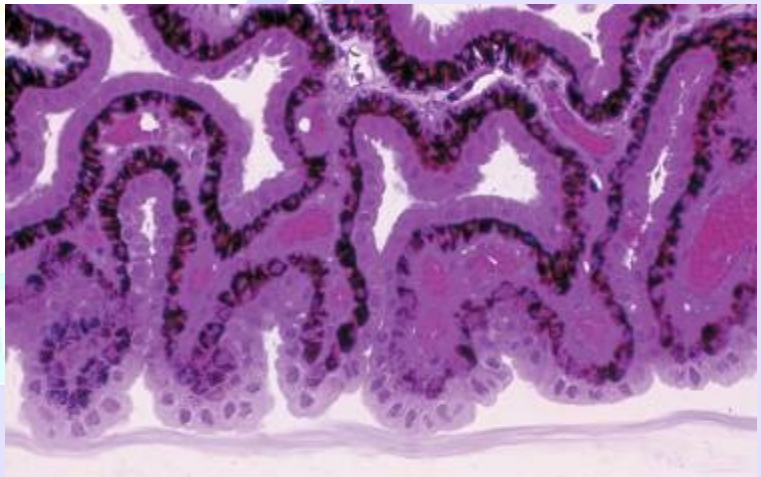


CILIARY BODY

- **1. Ciliary Muscle** forms bulk and consists of smooth muscle fibers, contraction of ciliary muscles causes relaxation of tension in the suspensory ligament of lens
- **2. Vessel Layer** (Ciliary stroma) contains loose connective tissue and blood vessel
- **3. Lamina Vitrea**
- A continuation of Bruch's membrane of choroid.



(C = cornea; I = iris; PE = iris pigment epithelium; CB = ciliary body; S = sclera; R = retina; RPE = retinal pigment epithelium; Ch = choroid.



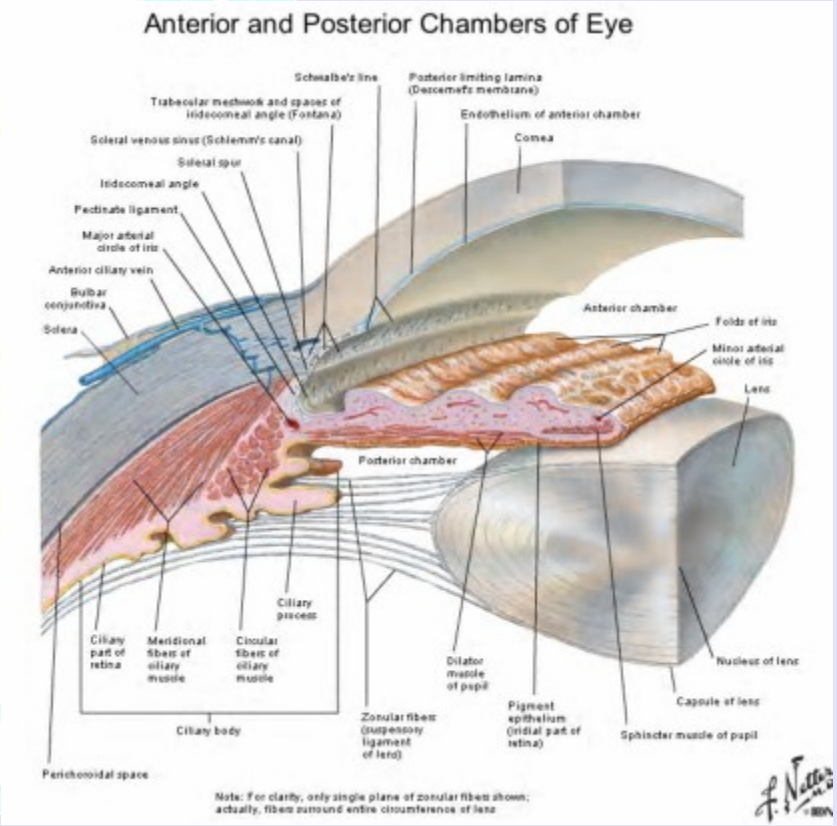
Section of ciliary processes showing their double layer of pigmented and nonpigmented epithelial cells. Note also the core of connective tissue

CILIARY BODY

- **4. Pigment Epithelium** is a continuation of pigmented epithelium of retina. Single layer of cuboidal cells. Cytoplasm of cells is heavily filled with brown black melanin granules. Summit of ciliary processes has much less pigment, so it is whitish.
- **5. Ciliary Epithelium** is forward continuation of sensory portion of retina is single layer of columnar unpigment cells. It secretes aqueous humor.
- **6. Internal Limiting Membrane** is a thin basal membrane of ciliary epithelium.

AQUEOUS HUMOR

- It is a watery fluid, has an inorganic ion composition similar to that of plasma, but contains less than 0.1% protein (plasma has about 7% protein)
- It has a high concentration of ascorbic acid, free amino acids, and sodium, chloride,& bicarbonate ions
- Glucose & urea levels are also lower than those of plasma



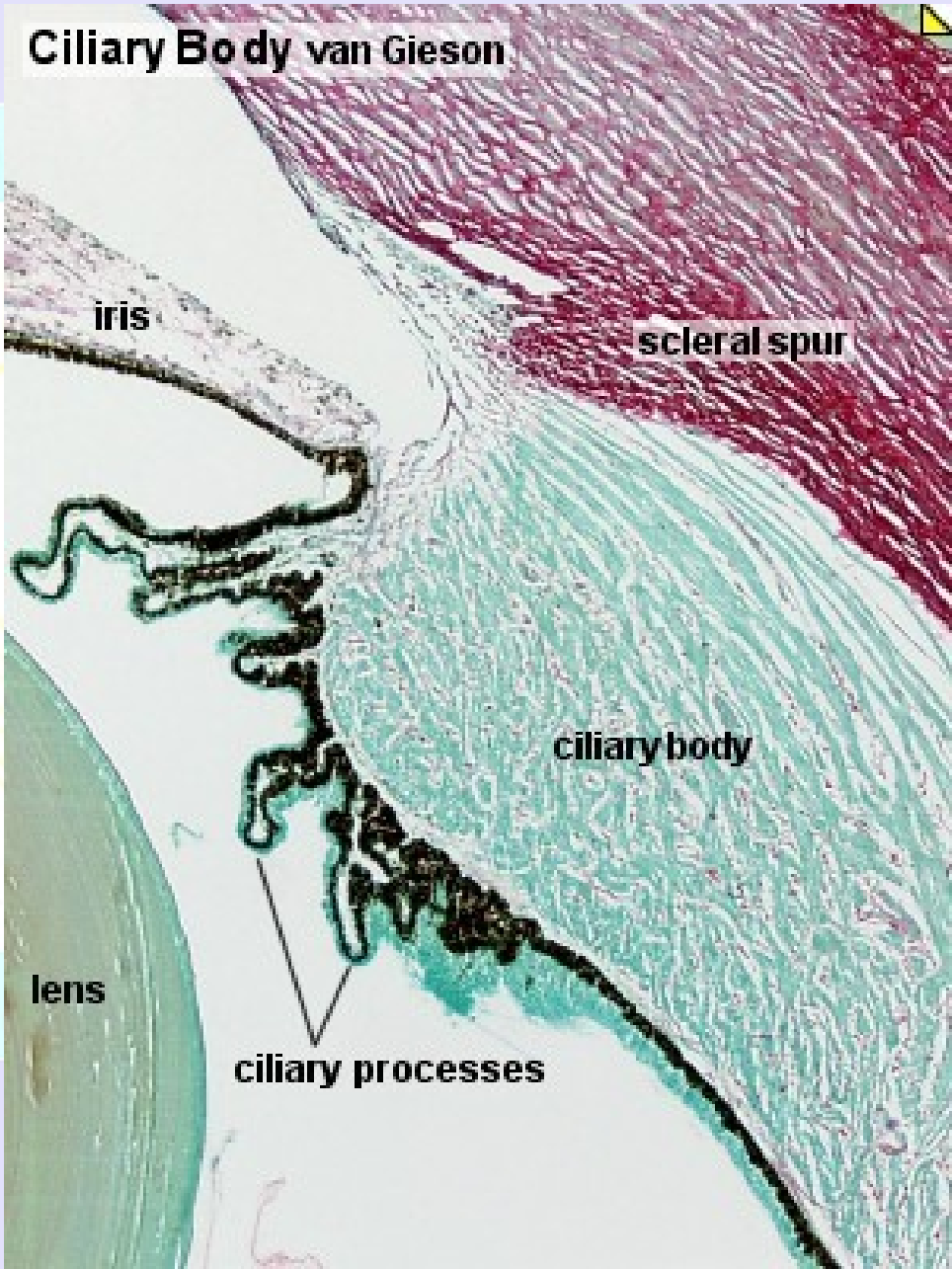
IRIS

- Microscopically consists 2 layers:

1. Iris stroma from mesenchyme.
2. Two epithelial layers posteriorly from neuroectoderm.

SECTION OF IRIS

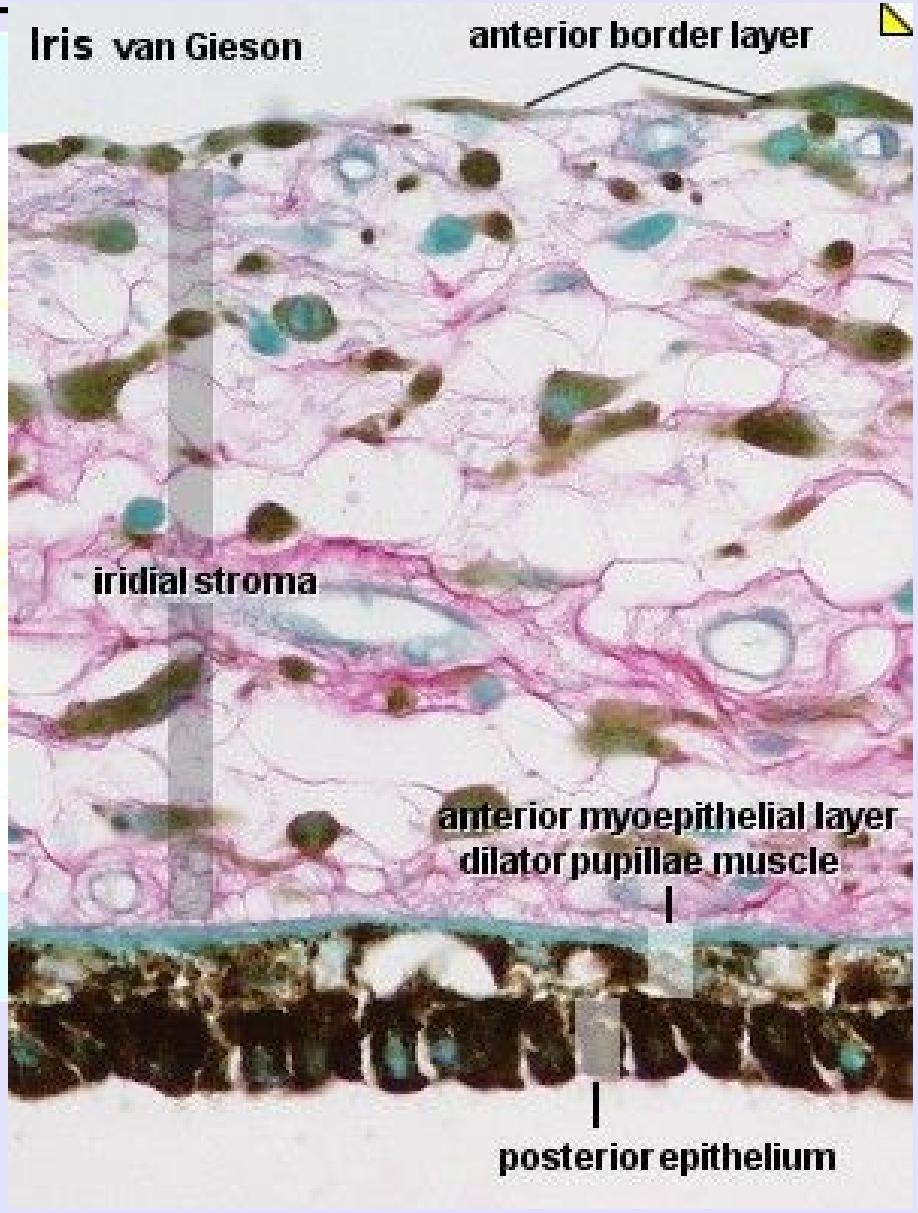
- A structure consisting of **a core of connective tissue** highly vascularized in certain regions
- The outer layer contains fibroblasts and very few pigmented cells
- In contrast, the **inner covering layer is heavily pigmented** to protect the eye's interior from stray light.
- **Dilator and constrictor (sphincter) pupillary** muscles control the diameter of the pupil.



SECTION OF IRIS

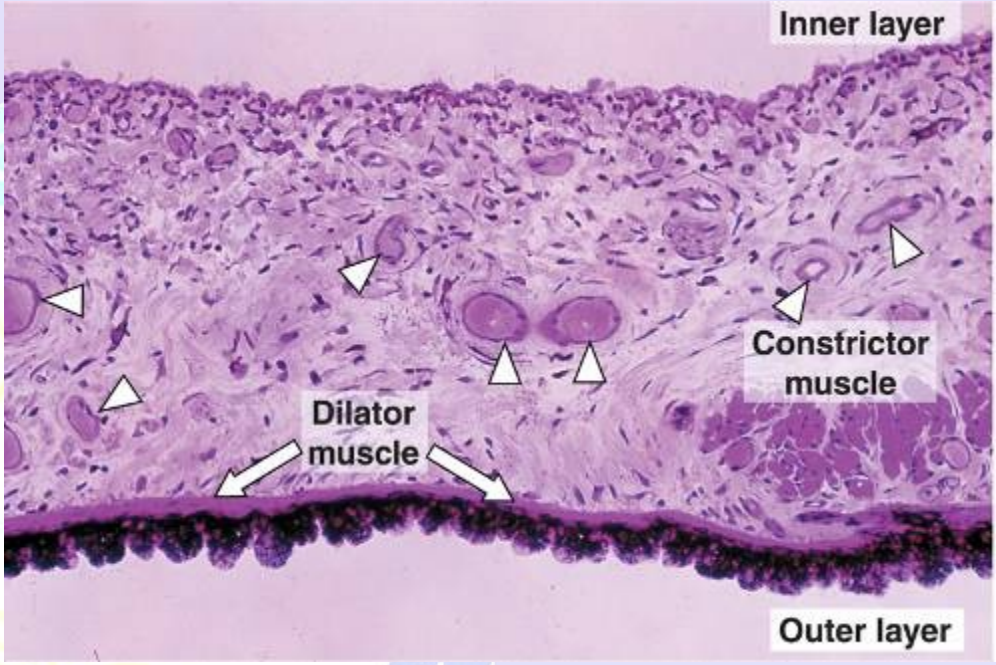
1. IRIS STROMA

- It contains connective tissue, blood vessels, nerve fibers, melanocytes, and smooth muscle of the sphincter pupillae and myo -epithelial cells of the dilator pupillae.
- **Colour of iris** depends on the **density of melanin granules present in the melanocytes of iris stroma**, more amount of melanin in melanocytes makes darker the colour of iris.



SECTION OF IRIS

- **Sphincter Pupillae Muscle**
- Smooth muscle fibers form a ring around the pupil, about 1 mm wide, when contracts – pupil constricts.
- N.S.- Parasympathetic postganglionic nerve fibers, from ciliary ganglion through the short ciliary nerves. **Dilator**

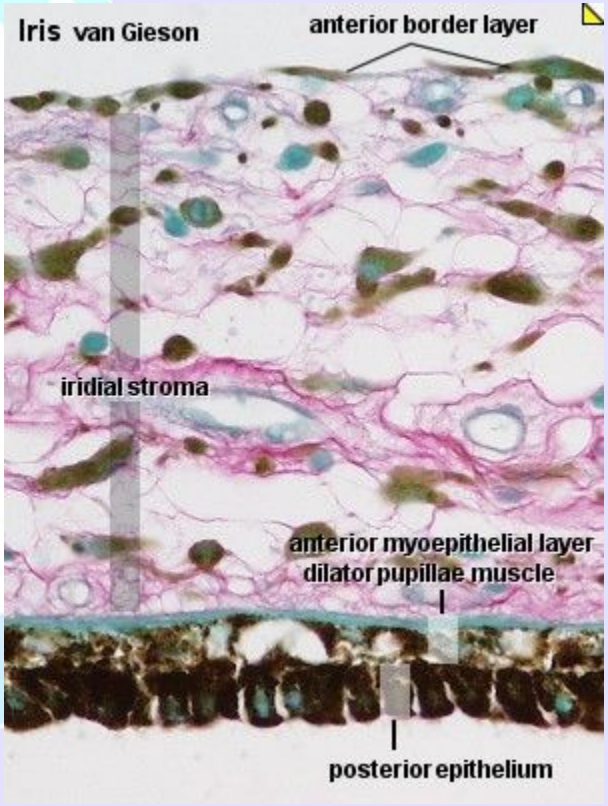


- **Pupillae Muscle**
- A thin sheet of myoepithelium which extends from iris root to the sphincter pupillae, when contracts – pupil enlarges.
- N.S. – Sympathetic postganglionic nerve via long ciliary nerves.

SECTION OF IRIS

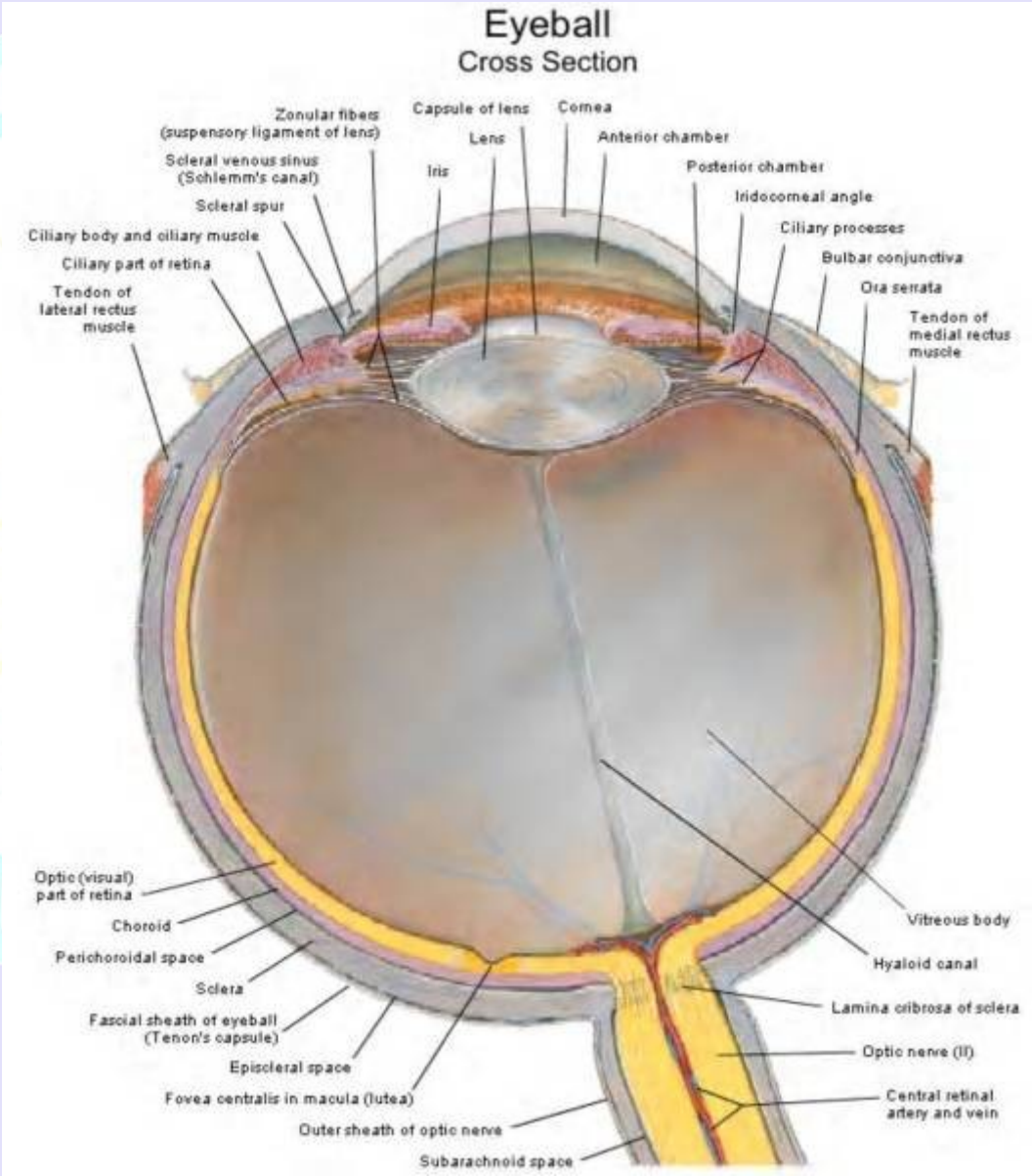
2. TWO POSTERIOR EPITHELIAL LAYERS OF IRIS

- Cells of both layers are cuboidal in shape & contain brownish black granules of melanin pigment.



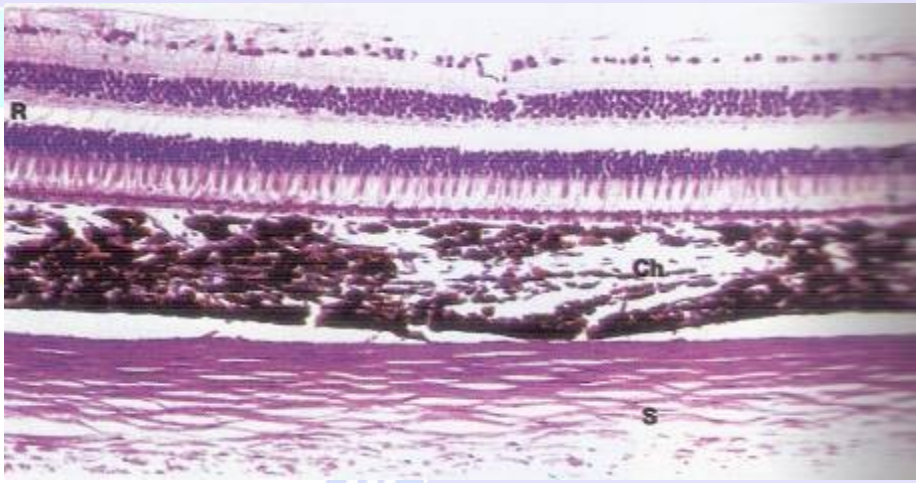
VITREOUS BODY

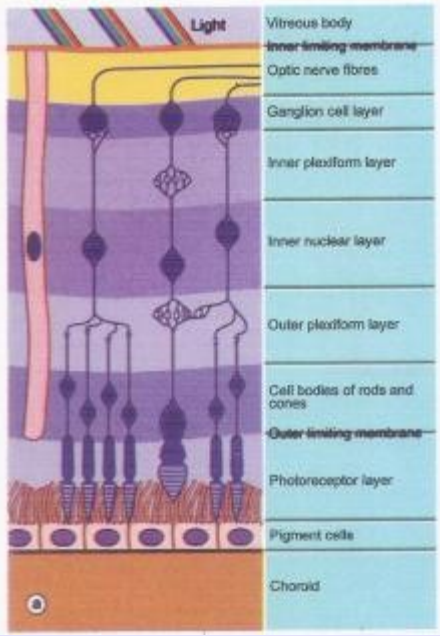
- Occupies the region of the eye behind the lens.
- A transparent gel that consists of water (about 99%),a small amount of collagen heavily hydrated with hyaluronic acid molecules and very few cells (hyalocytes).
- Hyalocytes synthesize collagen and hyaluronic acid.

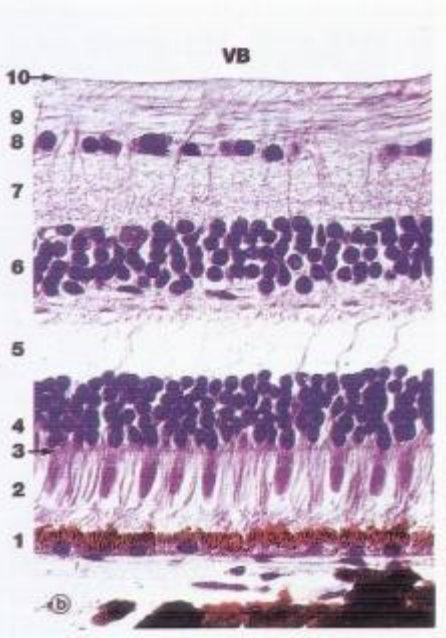


RETINA

- It is the innermost layer of the eyeball
- It transduces the stimulus of light into nerve impulses, resulting in the sensation of vision
- On the bases of structural features the retina is divided into **ten layers**



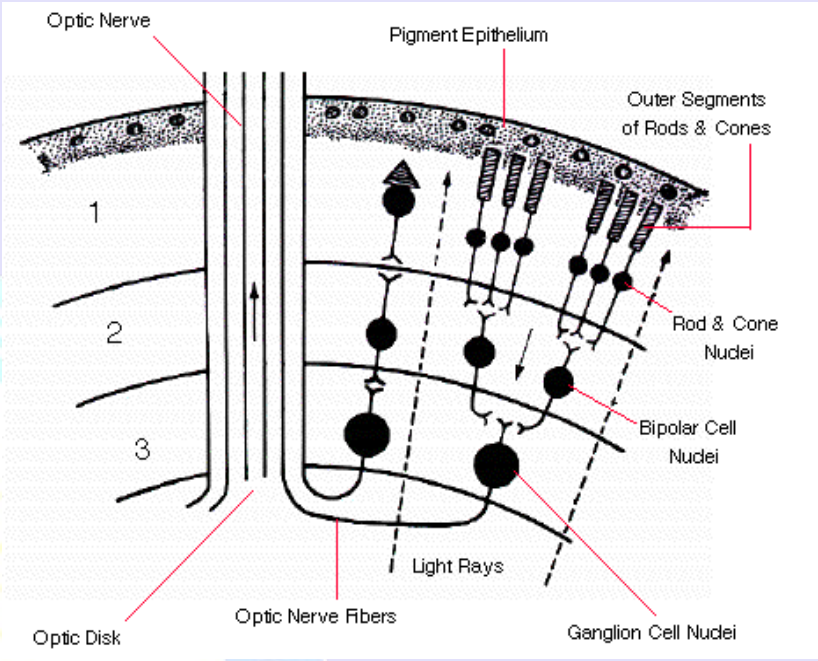


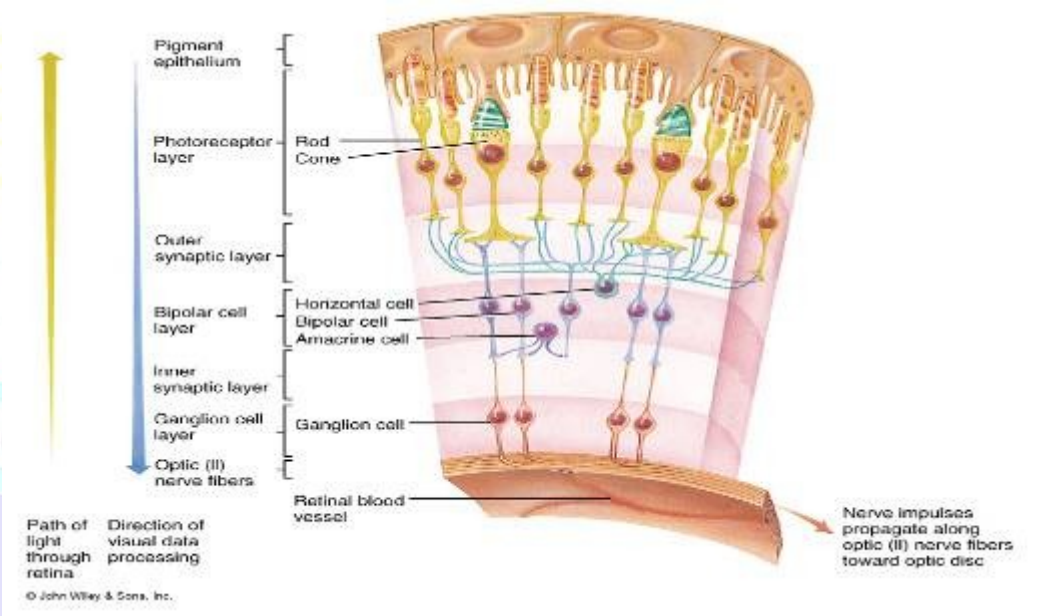


RETINA

- The true photoreceptors i.e. **rods and cones form the outermost layer** of neural retina, lying away from the light stimulus

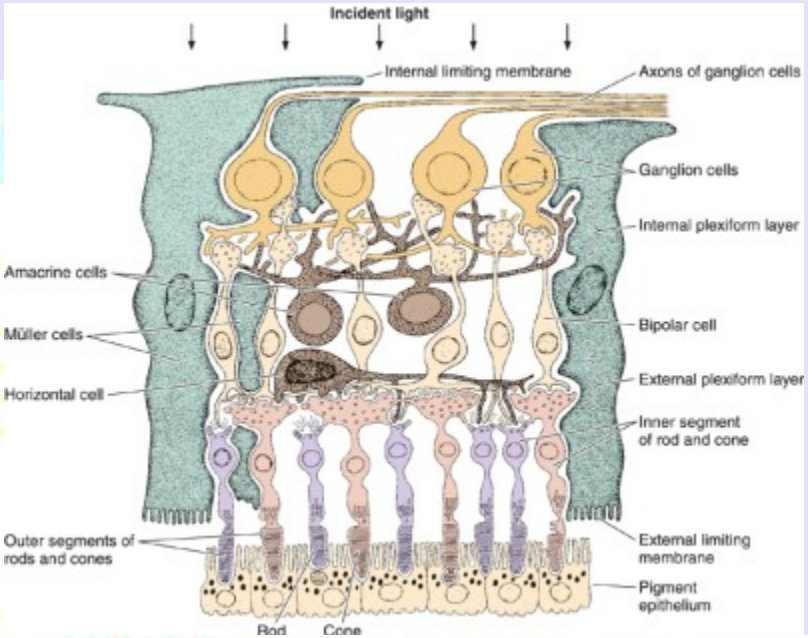
- Light must pass through the thickness of retina to reach the photoreceptors so **inverted retina** is a characteristic of all vertebrates

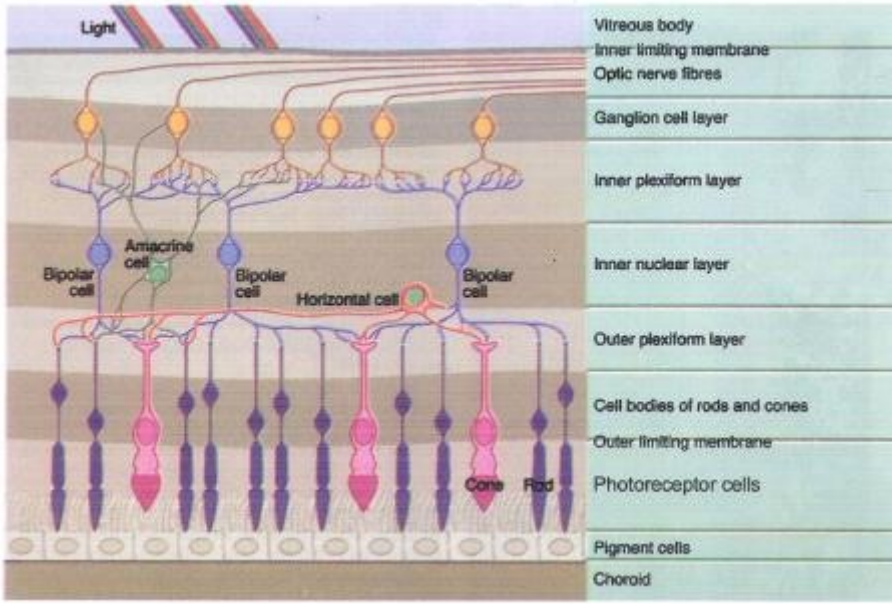




PIGMENT EPITHELIUM

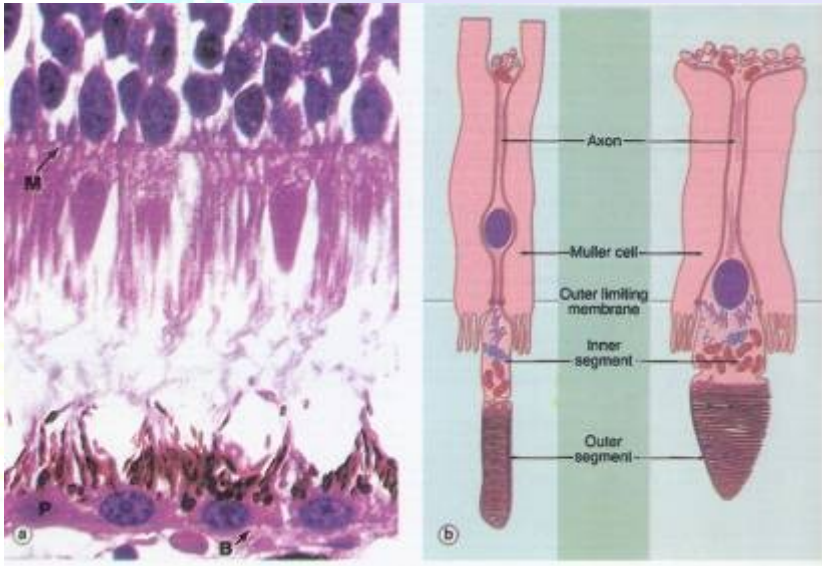
- It is a **single layer of low columnar cells** with basal nuclei
- Basal regions of cells adhere firmly to the lamina vitrea of choroid
- Adjacent cells are bound by junctional complexes
- **Apical part of each cell** has many cylindrical processes that surround and interdigitate with the outer segments of the rods and cones, but easily separable as occurs in detachment of retina.
- Apical cytoplasm of cells has numerous **melanin granules**





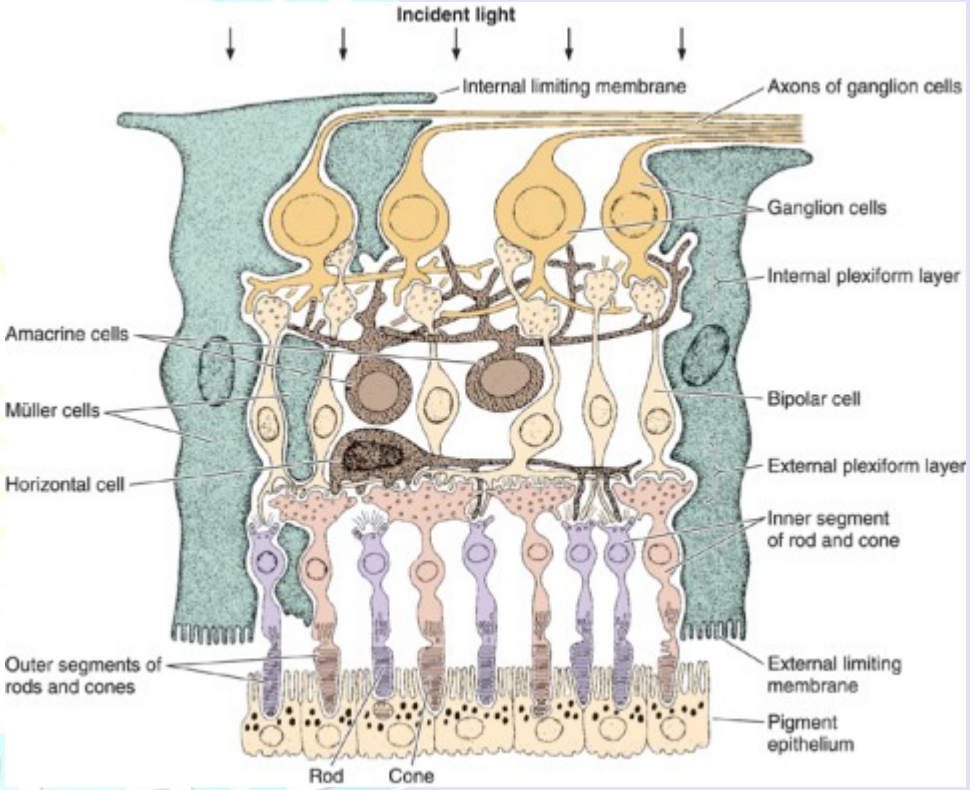
FUNCTIONS OF RETINAL PIGMENT EPITHELIUM

- **Absorption of light**
- **Storage of vitamin A** and its supply to the photoreceptors. Vitamin A is an essential precursor of the photosensitive pigments of rod and cone cells
- **Phagocytosis**
- Formation of **blood-retina barrier**



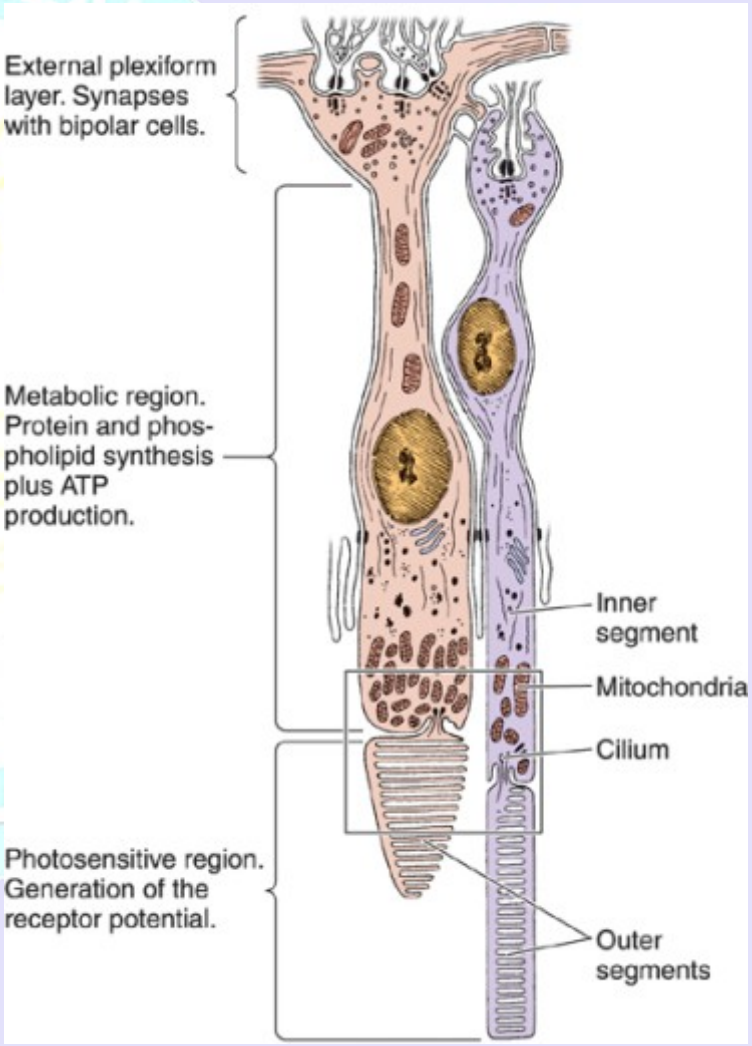
NEURAL RETINA

- It is only **three neurons deep**
- First neuron are rods and cones
- Second neuron are bipolar cells
- Third neuron ganglion cells



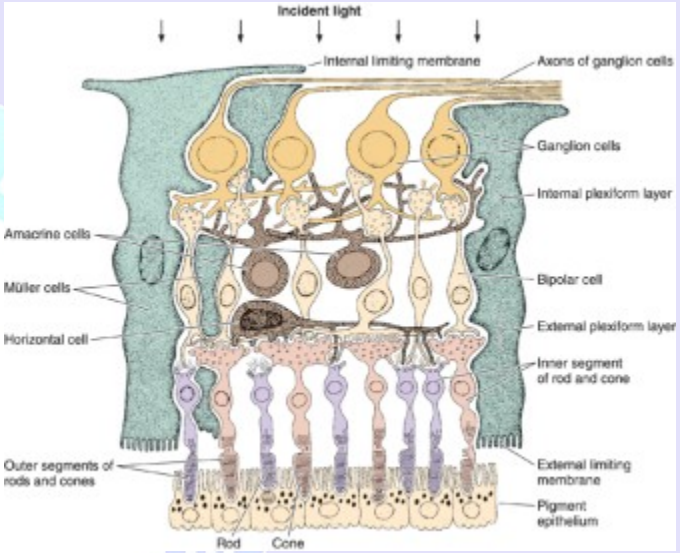
NEURAL RETINA

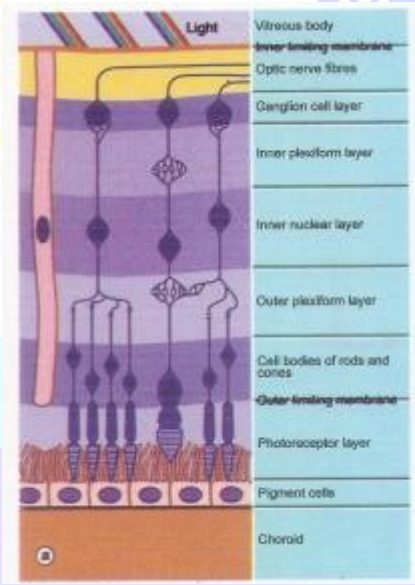
- **FIRST NEURON**
- Each rod and cone cell has **three parts**
- Sensory end organ (rod and cone) lies against pigment epithelium.
- Cell body contains nucleus.
- An inner terminal fiber.

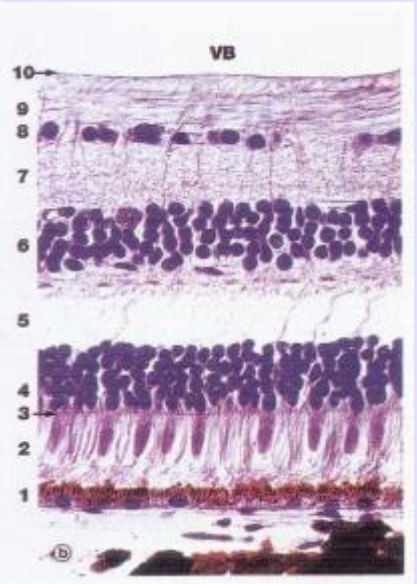


NEURAL RETINA

- **FIRST NEURON**
- Nuclei and cell bodies of rod and cone lie in the **outer nuclear layer** having **five rows of nuclei**
- Single outer row immediately next to the external limiting membrane is composed of cone nuclei, while remaining 4 rows consists of rod nuclei.
- At outer plexiform layer rods and cones make synaptic junctions with the dendrites of bipolar neurons.





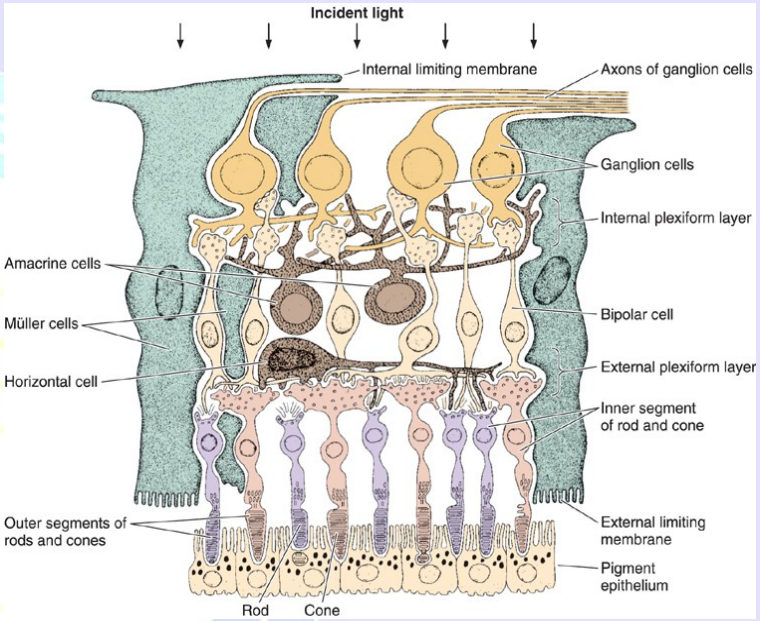


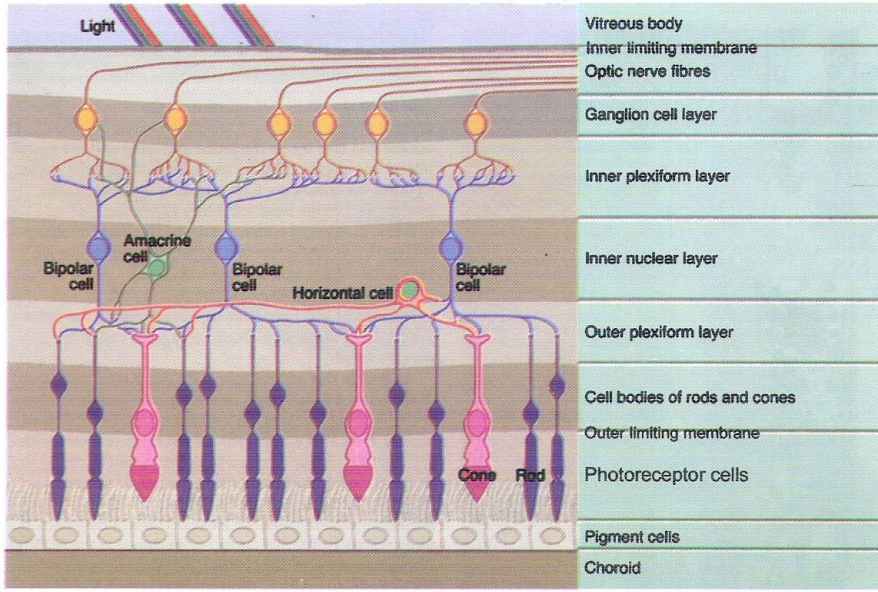
NEURAL RETINA

- **SECOND NEURON**

- **Bipolar cells** and their nuclei lie in the inner nuclear layer.

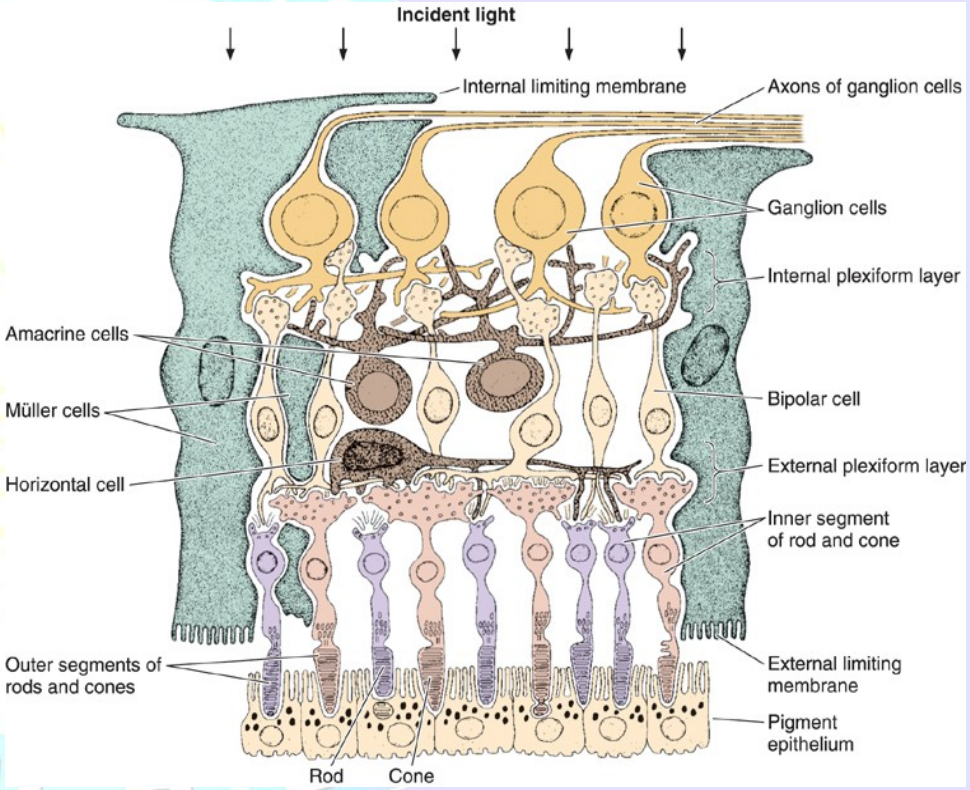
- Axons of bipolar neurons pass into the inner plexiform layer, where they synapse with the dendrites of the ganglion cells





NEURAL RETINA

- **THIRD NEURON**
- **Ganglion cells** and their large bodies lie in ganglion cell layer
- Long axons of ganglion cells course in the optic nerve fiber layer to the optic disc, where all the axons converge to form the optic nerve
- External and internal limiting membranes are not true membranes.
- **External limiting membrane** is region of junctional complexes between outer ends of retinal gliocytes (Muller's cells) and adjoining photoreceptor cells.
- **Internal limiting membrane** is formed by the apposition of expanded inner processes of Muller's cells and their basal lamina separating retina from vitreous body.

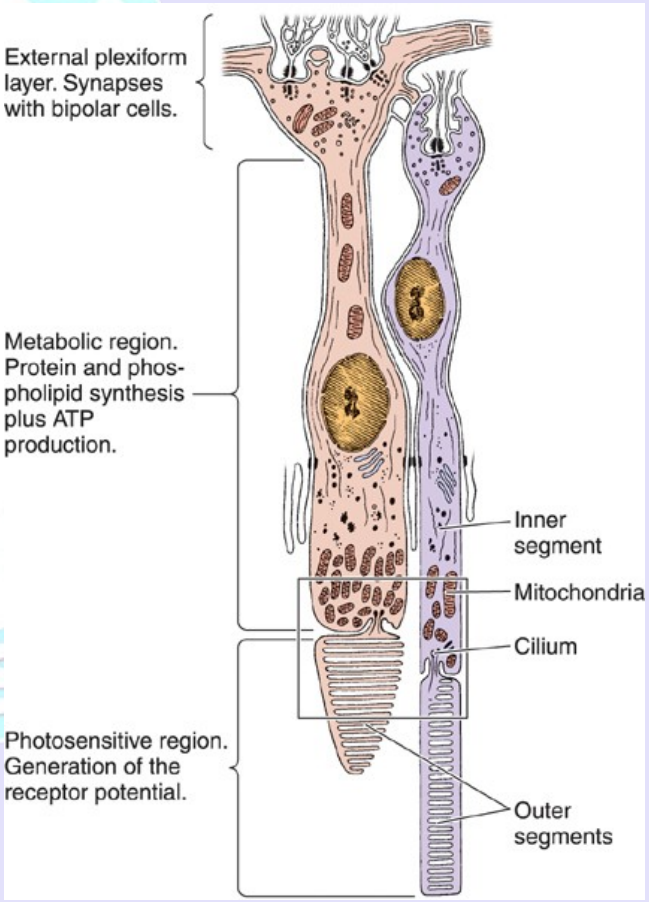


PHOTORECEPTORS

- Rods and cones are modified bipolar neurons.
- One pole is in the form of single photosensitive dendrite.
- Other is an axon, which makes synaptic junction with the dendrite of bipolar neurons.
- Bodies of rod and cone cells lie in the outer nuclear layer.
- According to the shapes of photosensitive dendrite of rod and cone, these cells are given name.

ROD CELLS

- Thin elongated cells with a cylindrical dendrite. Rod is divided into an outer segment 28µm long with diameter of 2µm, and an inner segment 32µm long with diameter of 2µm connected by eccentric narrow neck.
- **Total number is 110 to 125 million.**
- Outer segment shows transverse striations, and its tip is embedded.
- Responsible for sensing low levels of illumination and is indispensable for night vision.

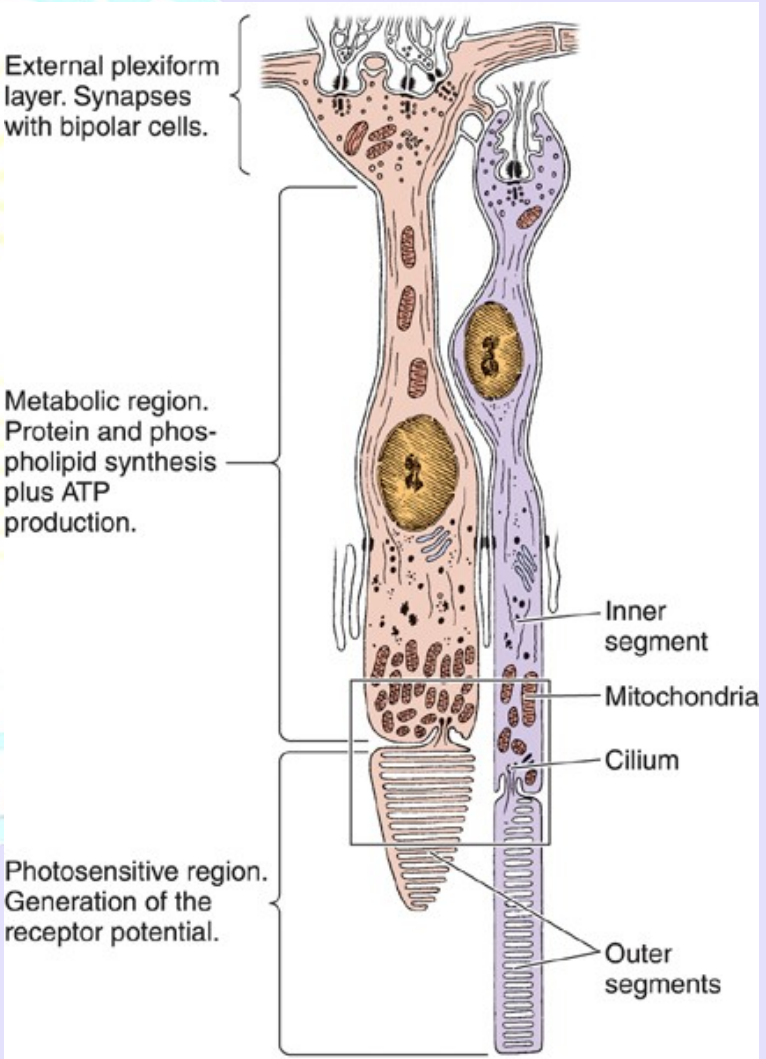


E/M OF ROD CELLS

- **Outer segment** **c**ontains 600 to 1000 flattened membranous lamellae or discs of uniform diameter, stacked like a pile of coins.

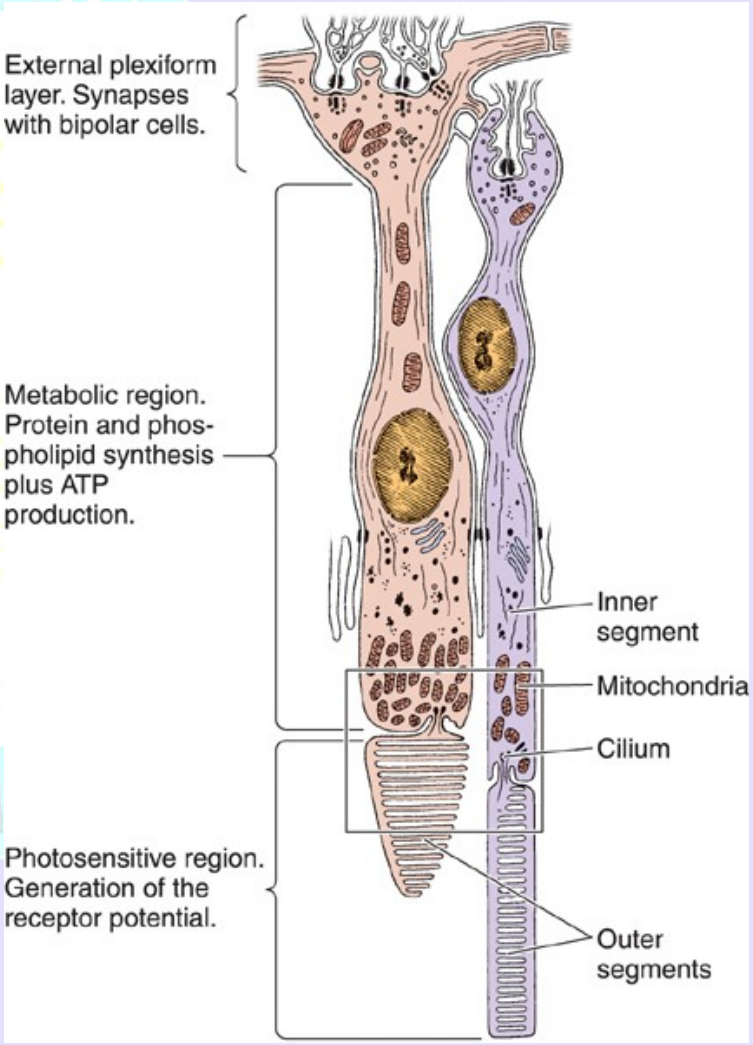
- Discs are separated from the cell membrane and contain the photosensitive pigment

- Neck contains axoneme of a cilium, which originates from a body located in the outer part of inner segment.



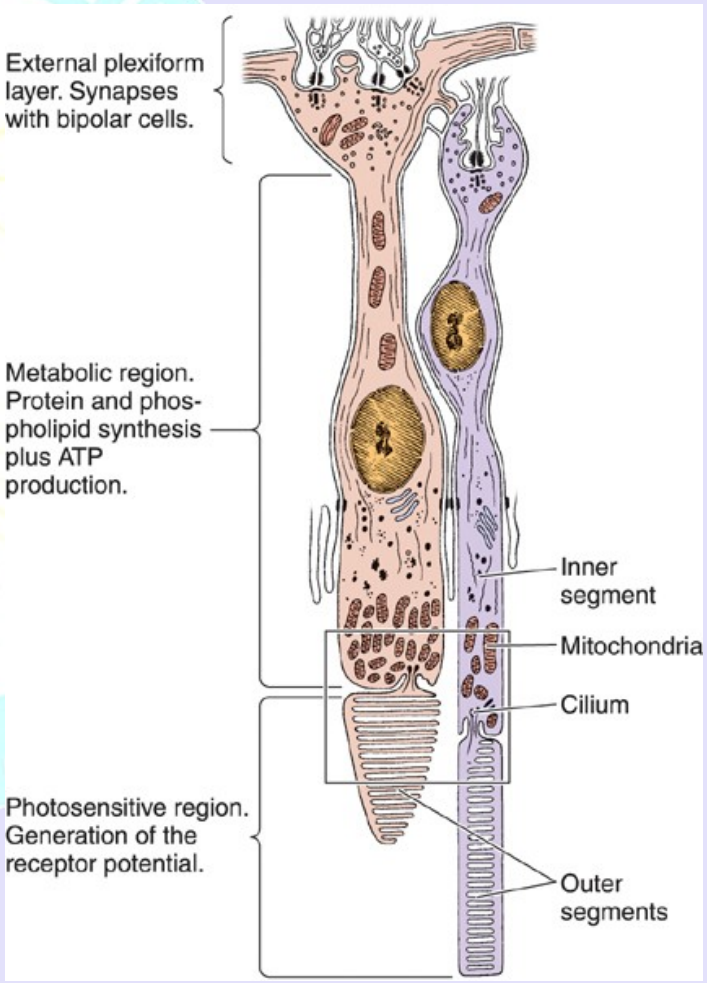
E/M OF ROD CELLS

- Inner segment is divided into two parts
- **Outer part** is ellipsoid and contains numerous mitochondria. **Inner part (myoid)** contains golgi apparatus, smooth and rough endoplasmic reticulum for protein synthesis
- Dendrite is connected to the body by a slender outer rod fiber that passes through external limiting membrane.
- An inner rod fiber passes from the body into outer plexiform layer to end in a small knob called **"rod spherule"** which makes synaptic contact with dendrites of bipolar and axons of Horizontal cells.



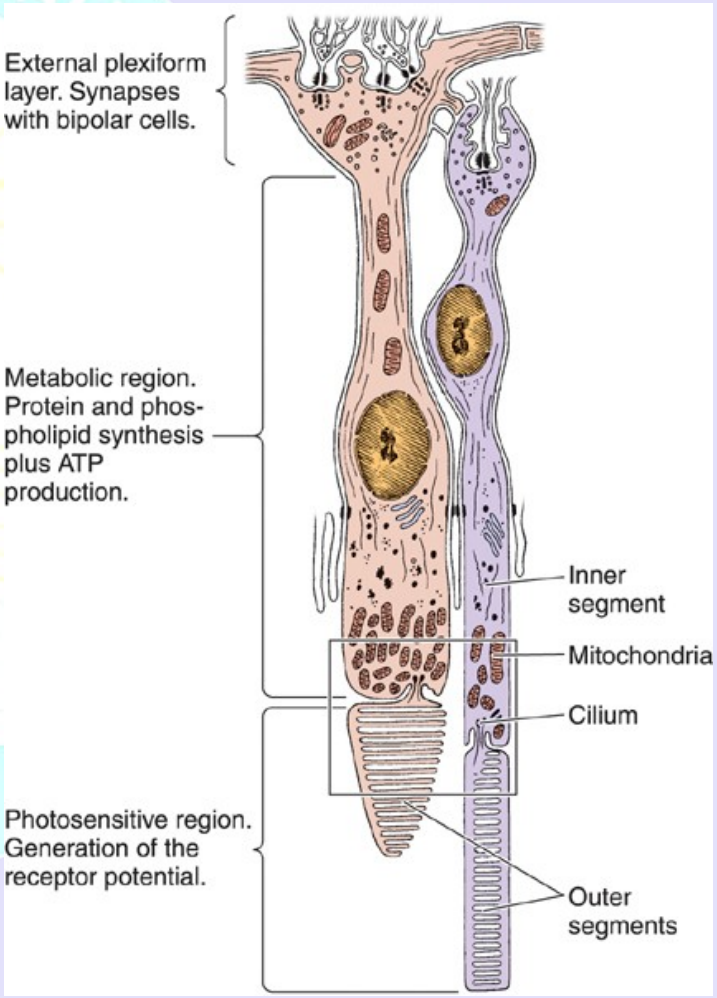
CONE CELLS

- Approximately **6 millions**
- Respond only to bright light and responsible for colour perception
- Cone shows a tapering outer and a conical inner segment Both are flask-shaped
- Stacked lamellae or discs of outer segments are not separated from the cell membrane, rather they represent repeated invagination of this membrane.



CONE CELLS

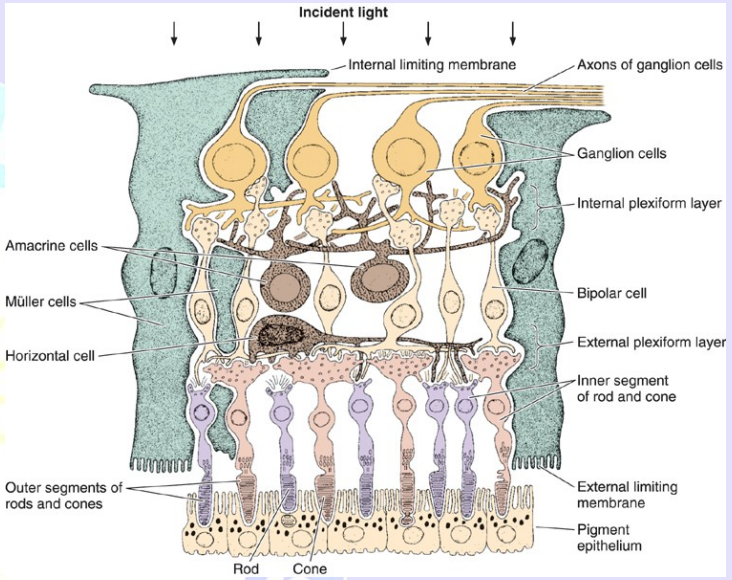
- **Nucleus is larger than rod cell** and lies just internal to external limiting membrane (outer fiber of cone is shorter)
- Inner fiber expends at its termination in the outer plexiform layer to form **“cone pedicle”** which synapses with dendrites of bipolar neurons.
- Membranous lamellae of outer segments contain visual pigments (retinal pigment and photopsins, a type of protein).
- One of three types of colour pigments are found in each of different cones, thus making the cones selectively sensitive to blue, green and red colour

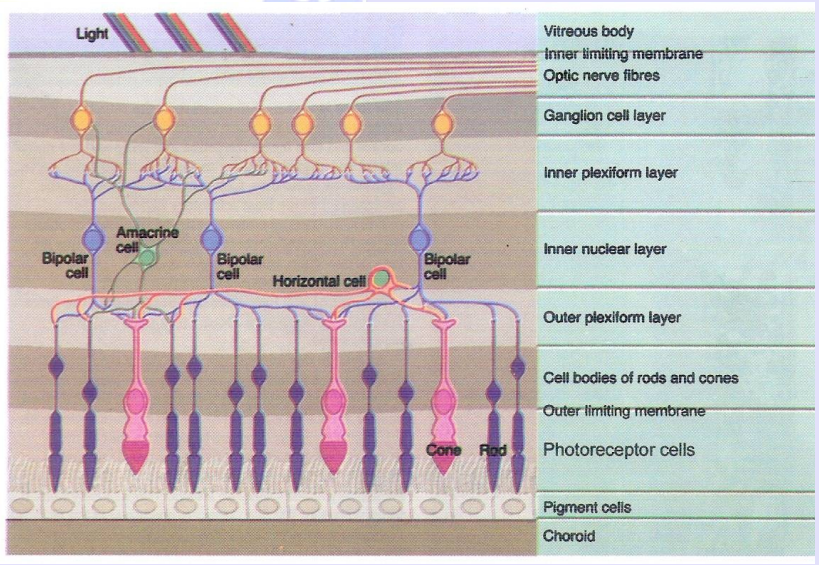


ASSOCIATED NEURONS

1. HORIZONTAL CELLS

- Bodies lie in the outer part of the inner nuclear layer
- Dendrites and axons lie in the inner part of the outer plexiform layer
- Dendrites terminate around numerous cone pedicles
- Single axon branches at its termination to synapse with rod spherules, as well as cone pedicles
- They connect a group of cone cells in one area with a group of rod and cone cells in another area



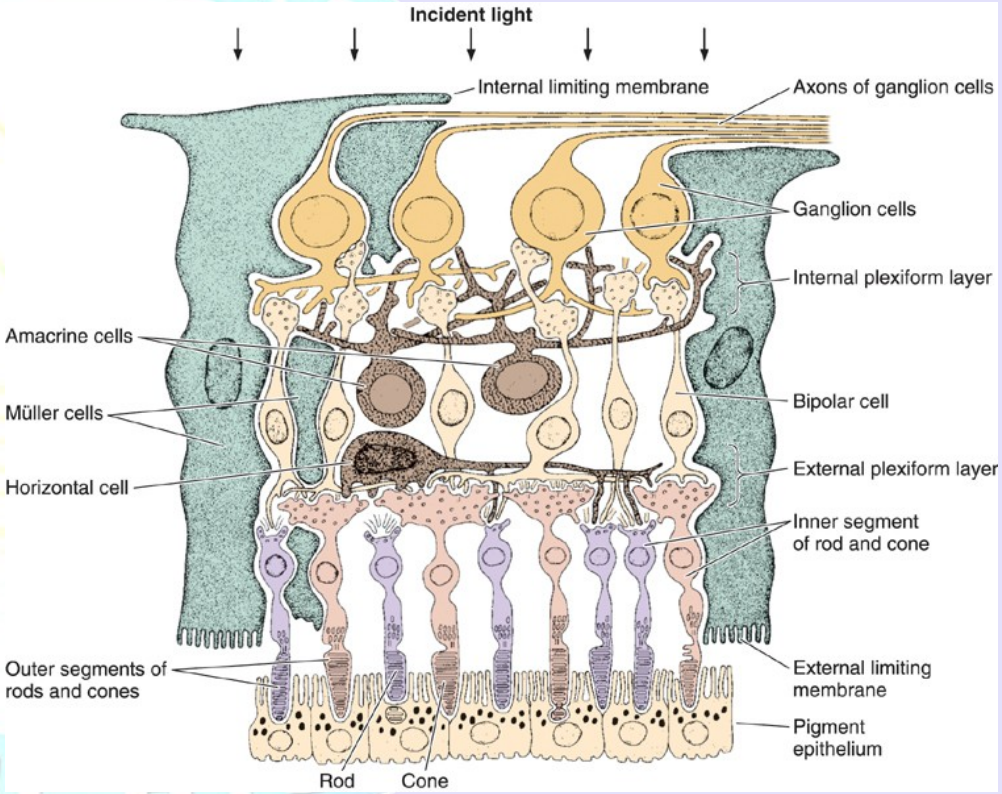


ASSOCIATED NEURONS

2. AMACRINE CELLS

- Cell bodies lie in the inner part of inner nuclear layer.

- Each cell has a single process which passes into the inner plexiform layer and extensively branch to synapse with the dendrites of several ganglion cells.



ASSOCIATED NEURONS

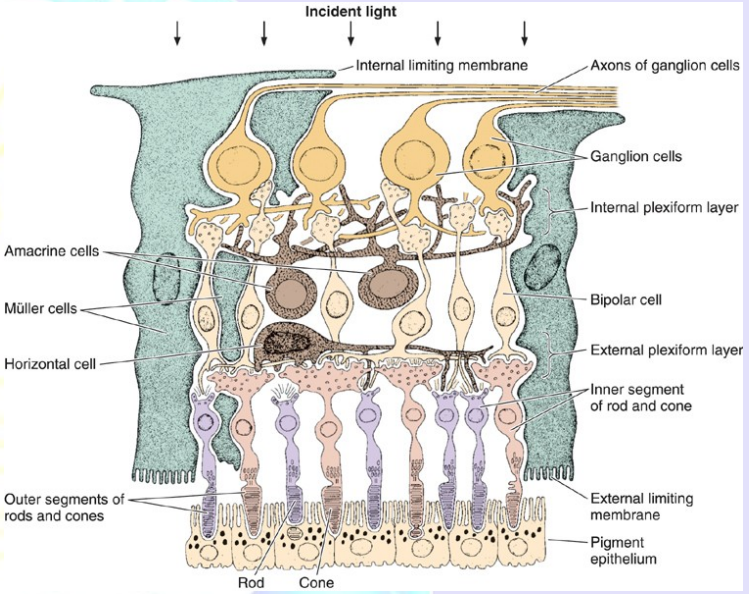
3. SUPPORTING ELEMENTS

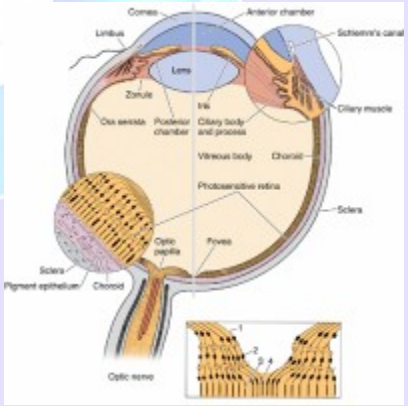
- **MULLER'S CELLS (RETINAL GLIOCYTES)**

- They have very large bodies, lie in the inner nuclear layer.

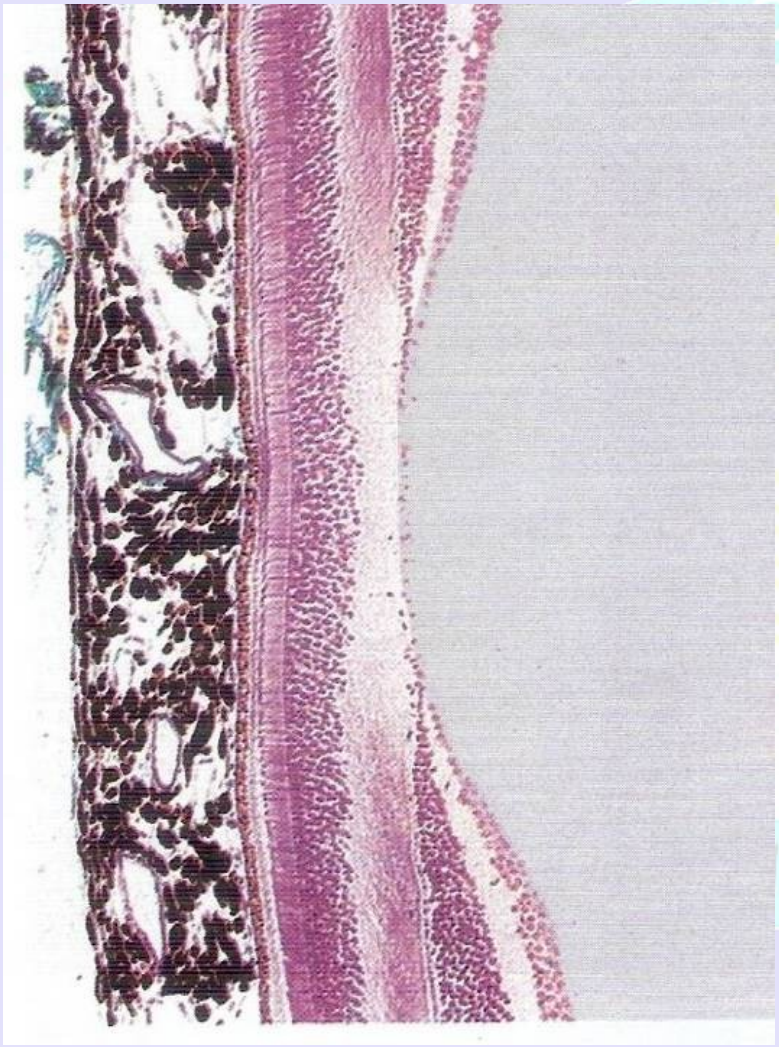
- Thin cytoplasmic processes extend from the cell body to the external and internal limiting membranes.

- They serve to **support**, **nourish** and **insulate** the retinal neurons and their processes.





FOVEA CENTRALIS



- A shallow, circular depression lying at the posterior pole of the optical axis of eye.
- Caused by the absence of inner layers of retina.
- Bipolar and ganglion cells accumulate in the periphery of depression, so that its center consists only of cone cells.
- In this area, blood vessels do not cross over the photosensitive cells.
- Light falls directly on cones in the central part of fovea, which helps account for extremely precise visual acuity of this region.

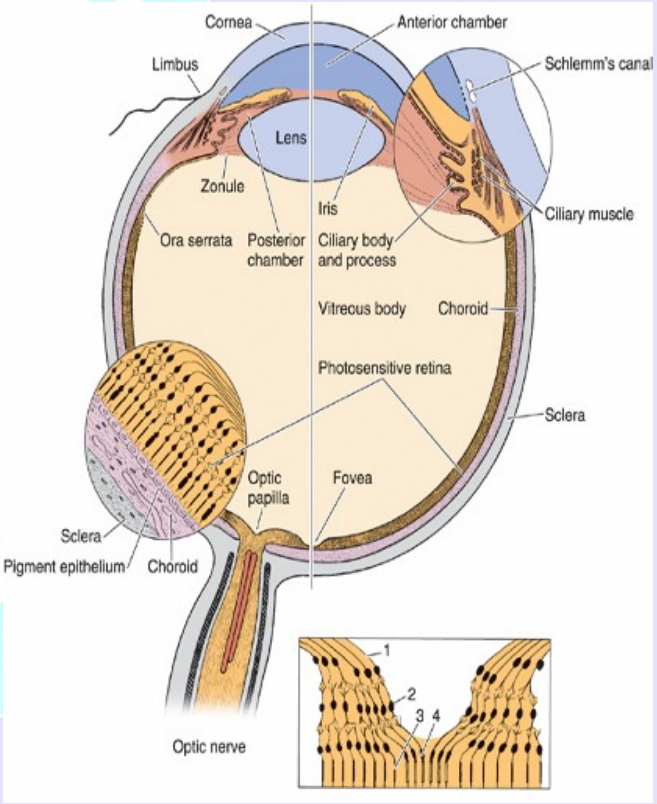
EYE LENS

- It is a transparent, biconvex body situated between the iris and vitreous body.

- It is elastic in the young subjects, and becoming harder with the age.

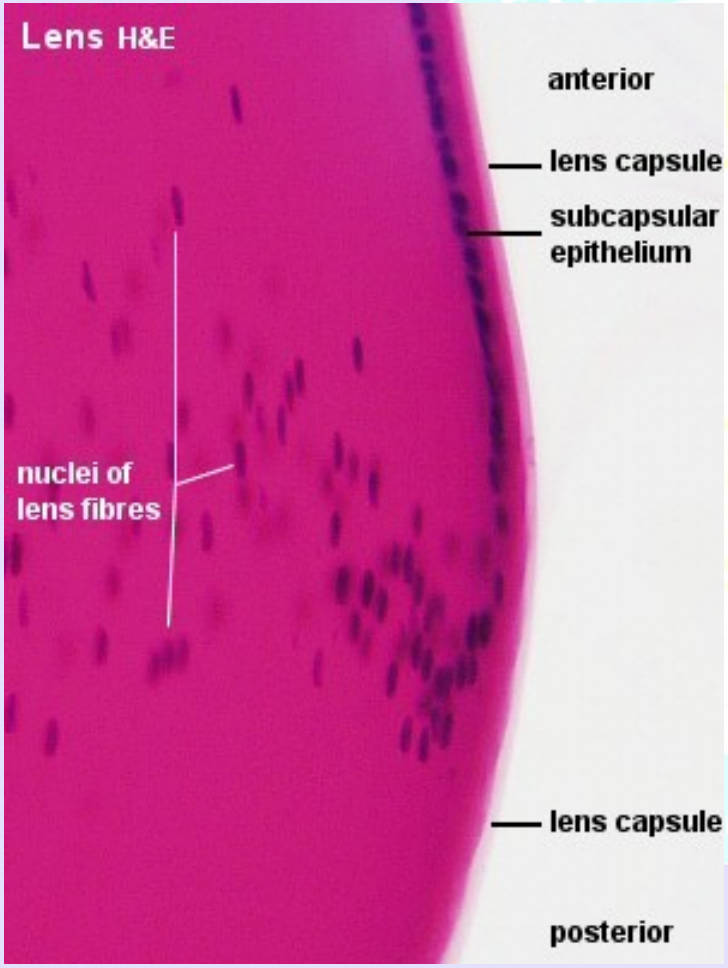
- **Structure:** It has three principal components

- Lens capsule.
- Subscapular epithelium.
- Lens substance.



EYE LENS

1. Lens capsule



- It envelops the lens
- It is 10 – 20 µm thick, homogeneous retractile membrane
- It is elastic
- It is composed of collagen type IV and glycoprotein

EYE LENS

2. Subcapular Epithelium

- It is a simple cuboidal epithelium, present only on the anterior surface of the lens just under the capsule.
- Towards the equator, epithelial cells increase in height and transform into lens fibers.
- Lens grows throughout life by the addition of these fibers

EYE LENS

3. Lens Substance

- It consists of elongated prismatic lens fibers.
- Lens fibers are highly differentiated cells, derived from cells of the subscapular epithelium.
- Lens fibers eventually lose their nuclei and other organelles and become greatly elongated, attaining dimensions of 7 – 10 mm in length, 8 – 10 µm in width, and 2 µm in thickness.



THE END